

Release Notes

Enlighten XO (Experience Optimization)

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Contents

Contents	1
Revision History	2
Enlighten XO (Experience Optimization)	3
Breaking Changes	3
Overview – What is Enlighten XO?	4
Enlighten XO – User Flow	4
Getting Started	4
Step 1: Create Enlighten XO Project	5
Step 2: Enlighten XO Processes Interactions	7
Step 3: Investigate Customer Intents	8
Step 4: Agent Tasks and Task Flows	11
Step 5: Add Default Custom Intent Phrases	13
Ancillary / Supporting Functionality	15
Object Management	15
Playback and Transcripts	16
Individual Interaction Task Flow	17
Data Export	17
Conversational flow	25
User-generated Flows	32
Bot Utterance Failure Analysis	36
User Activity Auditing	36
Enhanced Intent Trending	36
Intent Trending: Compare Metrics	37
Intent Trending: Compare Intents	38
Data Export via API	38
User Interface Localization & Accessibility	38
Compatibility	39
Known Issues/Behaviors and Workarounds	39
Key Resolved Issues	40
Appendix A: Summary of Features per Release	43

Revision History

Date	Version	Notes
July 31, 2022	Enlighten XO v1	Release Notes for Enlighten XO v1
September 30, 2022	Enlighten XO v12.4 FP2	Updated for Enlighten XO functionality added in conjunction with Nexidia v12.4 Feature Pack 2
November 30, 2022	Enlighten XO v12.4 FP3	Updated for Enlighten XO functionality added in conjunction with Nexidia v12.4 Feature Pack 3
March 30, 2023	Enlighten XO v12.4 FP4	Updated for Enlighten XO functionality added in conjunction with Nexidia v12.4 Feature Pack 4
May 31, 2023	Enlighten XO v12.4 FP4.2	Updated for Enlighten XO functionality added in v12.4 Feature Pack 4.2 Updated images to reflect new Intent Analysis visuals for Level 1, Level 2, and Level 3 Intents: 
August 2023	Enlighten XO v12.4 FP5	Updated for Enlighten XO functionality added in v12.4 Feature Pack 5.
September 2023	Enlighten XO v12.4 FP5.1	Updated for Enlighten XO functionality added in v12.4 Feature Pack 5.1.
October 13, 2023	Enlighten XO v12.4 FP5.1.1	Updated for Enlighten XO functionality added in v12.4 Feature Pack 5.1 x1
November 15, 2023	Enlighten XO v12.4 FP5.1.2	Updated for Enlighten XO functionality added in v12.4 Feature Pack 5.X x2
December 15, 2023	Enlighten XO v12.4 FP5.1.3	Updated for Enlighten XO functionality added in v12.4 Feature Pack 5.X x3
January 22, 2024	Enlighten XO v12.4 FP5.2.4	Updated for Enlighten XO functionality added in v12.4 Feature Pack 5.X x4
March 15, 2024	Enlighten XO v12.4 FP5.2.5	Updated for Enlighten XO functionality added in v12.4 Feature Pack 5.X x5
April 24, 2024	Enlighten XO v12.4 FP5.2.6	Updated for Enlighten XO functionality added in v12.4 Feature Pack 5.X x6
June 28, 2024	Enlighten XO v12.4 FP5.2.7	Updated for Enlighten XO functionality added in v12.4 Feature Pack 5.X x7
September 25, 2024	Enlighten XO v12.4 FP5.2.8	Updated for Enlighten XO functionality added in v12.4 Feature Pack 5.X x8
February 1, 2025	Enlighten XO v12.5 X9	Updated for Enlighten XO functionality added in v12.5 x9
July 21, 2025	Enlighten XO v12.5 FP1 X10	Updated for Enlighten XO functionality added in v12.5 Feature Pack 5.X x10

Enlighten XO (Experience Optimization)

Breaking Changes

v12.4 FP5

In v12.4 FP5, Enlighten XO added configurations for the location of the Predictive Classifier Models (audio and chat). The application delivers a model by default. Should the customer have custom model/s, those must be placed in an accessible location (shared folder), and the configuration values must be updated with the selected location/s after upgrade to v12.4 FP5. (See Step 1: Create Enlighten XO Project, below, for more information.)

Upgrading users (upgrading to v12.4 FP5) are recommended to clear their browser cache post-upgrade.

v12.4 FP5.1

Users upgrading from v12.4 FP5 to v12.4 FP5.1.x should be aware that Entities previously detected (using v12.4 FP5) will not be present in v12.4 FP5.1.x due to a change in the way the location of those Entities is determined. For those projects, users can proceed to use them without Entities, or can re-create the project to identify Entities using the new method.

v12.4 FP5.2 X8

In v12.4 FP5.2 X8, Enlighten XO supports exporting (via “AutoFlow”) intents and conversational flows formatted for any supported bots. This is done via configuration parameter (Nexidia.XO.Bot) as part of the NIA Management Console.

In v12.4 FP5.2 X8, export to BPMN, is no longer available.

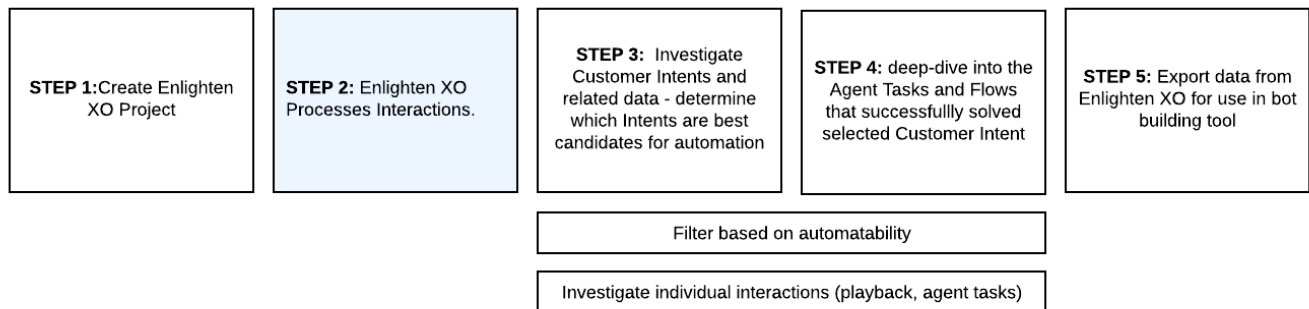
Overview – What is Enlighten XO?

Over the past several years, people have become generally more accustomed and willing to utilize self-service technologies rather than interacting with a live agent via call or chat in their dealings with corporate organizations. Often, this self-service takes the form of a voice bot or chat bot. However, the self-service experience often comes up short in terms of providing an excellent customer service experience, with customers too frequently abandoning (voluntarily or otherwise) the self-service channel and ending up interacting with a live agent to complete their intended task. Human conversation is complex. Anticipating the different paths that conversations may follow to complete a given customer intent has proven quite difficult, often involving guess work about how interactions would be expected to flow.

Enlighten XO helps self-service organizations build better self-service (bots), by analyzing data from actual agent-involved interactions and serving that up to bot builders so they can understand how those interactions *actually did* flow. Bot builders can program their self-service products to flow based on actual, proven successful interactions, reducing time to program the bot, improving likelihood of containment in the self-service channel, and increasing customer satisfaction.

Enlighten XO – User Flow

The typical user flow for Enlighten XO can be thought of as follows. Each of these steps will be described in the remainder of this document:



Getting Started

Enlighten XO comes out of the box with two standard User Profiles that enable customers to rapidly configure their application users with appropriate privileges in the system.

- XO Consumer - Enlighten XO Consumer users are standard users who can view and interact with the results of a given Enlighten XO project. These users can view data for Customer Intents, Agent Tasks, and export data from the system.
- XO Admin - XO admin would have full permission in the Enlighten XO applications. They have all the same privileges that XO Consumer users have, plus privileges to create/edit/delete projects, configure ROI calculation parameters and view results, and more. These users also have privileges for user and media management, allowing them to take the steps necessary to fully administer the Enlighten XO application.

Enlighten XO customers can edit these out of the box Profiles as they wish and/or create new ones to meet their specific governance needs.

Step 1: Create Enlighten XO Project

NOTE - There are 2 new configuration options in the Management Console as of v12.4 FP5, `Nexidia.XO.AudioPredictiveClassifierModel.SharedPath` and `Nexidia.XO.ChatPredictiveClassifierModel.SharedPath`, to store the “.pred” model files at a specified location so during the Enlighten XO upgrade process, they will not be overwritten with the default models. These are the models that determine automatability for individual interactions within an Enlighten XO project.

Users with appropriate privileges can create and activate an Enlighten XO “project”, which will result in the Enlighten XO engine processing media configured in the project definition. Users within the system can view Enlighten XO projects whose media they have Access Rights to (specifically, projects that use Target Media Sets whose reference Sessions they have access to), allowing customers to maintain control over which users can access which interactions.

XO Project						Create Project
Start a capture						
NEW PROJECTS (0)						
PROJECT NAME	PROGRESS	TARGET MEDIA SET	MODEL	CREATION DATE	CREATED BY	
No data to display						
FINAL PROCESSING (0)						
PROJECT NAME	PROGRESS	TARGET MEDIA SET	MODEL	CREATION DATE	CREATED BY	
No data to display						
AUTOMATION (1)						
Demo Synthetic 5	Done	XO Synthetic	DefaultIntents.ipm	Jul 29, 2022 5:43 PM	DA Default Administrator	

To create an Enlighten XO Project, the user *must*:

1. Have the “Enlighten XO – Projects Create” privilege, which controls availability of the “Create Project” button
2. Give the Project a name (must be unique within the Enlighten XO instance)
3. Select a Target Media Set (user will only be able to see/select Target Media Sets that reference Sessions that this user has Access Rights to)
4. Upload an Enlighten XO model (.ipm) file by Browsing or dropping, using the options shown here.

Starting from X7, dual topic intent model is supported.

Starting from X8, the end user can select an intent model from a pre-defined common folder on the server, located in the shared Enlighten XO directory (“Media\EnlightenXO\All Intent Model”) instead of uploading it directly. Most users would not likely have direct access to the Media folder and thus would upload an .ipm file using the ‘browse or drop’ control on this page. The system will then add that so that the .ipm file will be selectable from the list going forward.

Create Project

PROJECT NAME *
Enter project name

DESCRIPTION
Enter project description

TARGET MEDIA SET *
Select TMS

SELECT OR UPLOAD MODEL *
Select from list

OR

Drag files here or [Browse](#)
Supported file type: IPM

ACTIVATE WHEN READY ☐

ACTIVATE TRENDING REPORT ☐

Cancel Save

In addition, the user may:

- Provide a Description for the project
- Activate the project (the project will not process interactions unless/until it is activated)
- Activate the project for Intent Trending
- Add Intent Descriptions. Will be required to use Automatic Conversational Flow Builder

Users with appropriate privileges can configure the project by entering cost data for audio (cost per minute / per call) and/or cost per chat. Further, Target Containment (%) allows customer to indicate how much of the potential ROI they feel that they are likely to realize.

Once configured, users with privilege to view ROI data will see ROI metrics for intents and tasks within the project.

COST CALCULATION

Demo Synthetic 5
XO Synthetic
DefaultIntents.ipm
Project creation date: Jul 29 2022

CALL COST
\$ 20 PER: minute

CHAT COST
\$ 50 PER CHAT

TARGET CONTAINMENT
100 %

Enlighten XO Integrated with Interaction Analytics

For instances of Enlighten XO with Interaction Analytics, users with appropriate privilege will use the Discovery > Media & Filters > Media Set Administration page to configure Target Media Sets for transcription and indicate the appropriate Phrase Pack for each. Those that have been configured as such will be eligible for use in Enlighten XO projects.

Enlighten XO Standalone Implementation

For instances of Enlighten XO as a standalone application, all Target Media Sets in the system will automatically have 100% of their audio interactions transcribed. The Phrase Pack to use **must be indicated** in Management Console, via the “Nexidia.XO.Only.PhrasePackName” configuration.

Step 2: Enlighten XO Processes Interactions

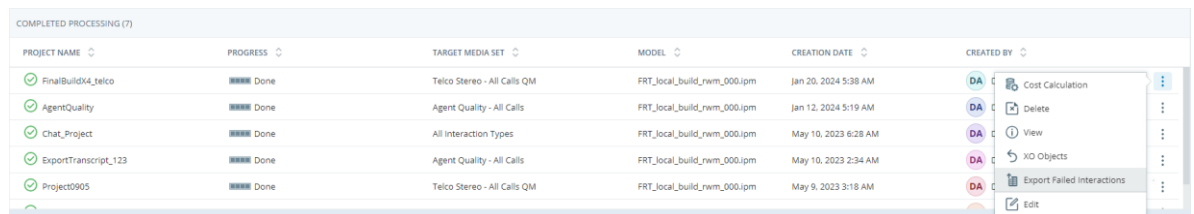
Once the Enlighten XO project has been created and activated, the system will process interactions from the media set (default maximum of 500,000, configurable) to:

- Identify the primary customer Intent for each
- Determine whether each is a good example of an Automatable interaction
- Compile key utterances from the live interactions for each Customer Intent and Agent Task

When the project has completed processing, the user can view results via the Enlighten XO User Interface.

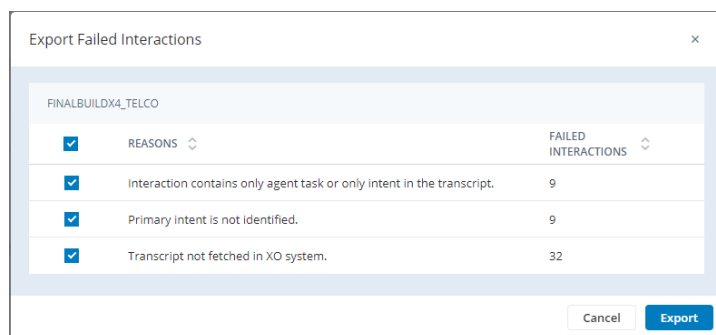
Starting with v12.4 FP5.2 X4, within the Project Administration area, users can view the interactions that are not included in the Project, and the reasons.

- The context menu of the processed project contains a new option (pictured below). User must have the “Enlighten XO – Export Data” privilege to see this option:



PROJECT NAME	PROGRESS	TARGET MEDIA SET	MODEL	CREATION DATE	CREATED BY
FinalBuildX4_telco	Done	Telco Stereo - All Calls QM	FRT_local_build_nwm_000.ipm	Jan 20, 2024 5:38 AM	DA
AgentQuality	Done	Agent Quality - All Calls	FRT_local_build_nwm_000.ipm	Jan 12, 2024 5:19 AM	DA
Chat_Project	Done	All Interaction Types	FRT_local_build_nwm_000.ipm	May 10, 2023 6:28 AM	DA
ExportTranscript_123	Done	Agent Quality - All Calls	FRT_local_build_nwm_000.ipm	May 10, 2023 2:34 AM	DA
Project0905	Done	Telco Stereo - All Calls QM	FRT_local_build_nwm_000.ipm	May 9, 2023 3:18 AM	DA

- The “Export Failed Interactions” dialog presents the failure reasons and count of interactions for each. Clicking “Export” downloads these details in a CSV file. The downloaded CSV file includes details of each interaction: name, interaction ID, and reason for failure.



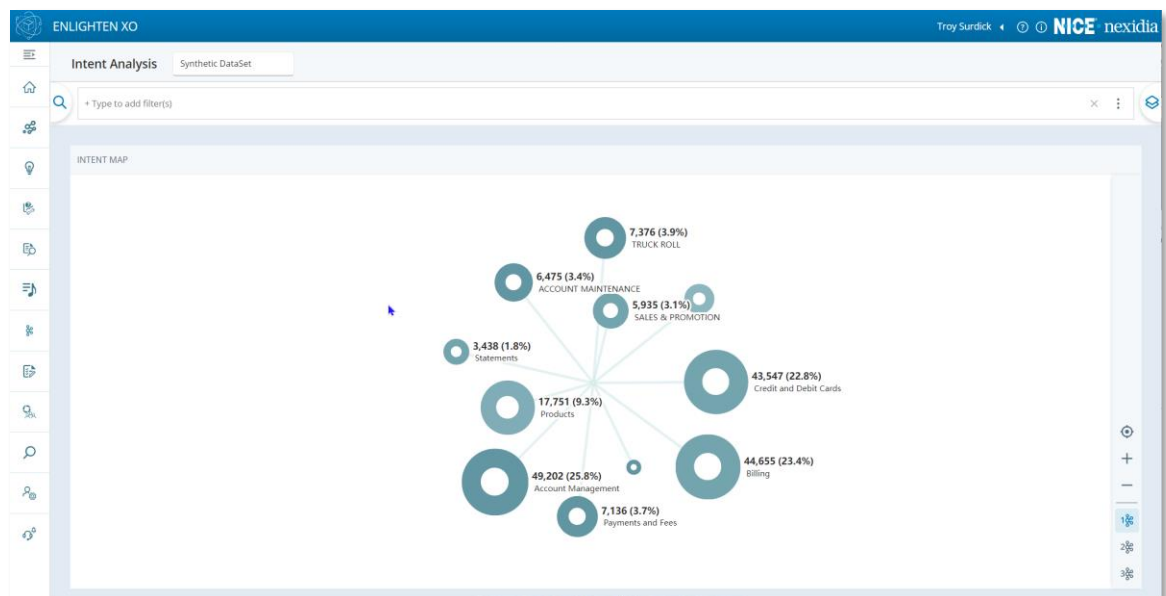
REASONS	FAILED INTERACTIONS
Interaction contains only agent task or only intent in the transcript.	9
Primary intent is not identified.	9
Transcript not fetched in XO system.	32

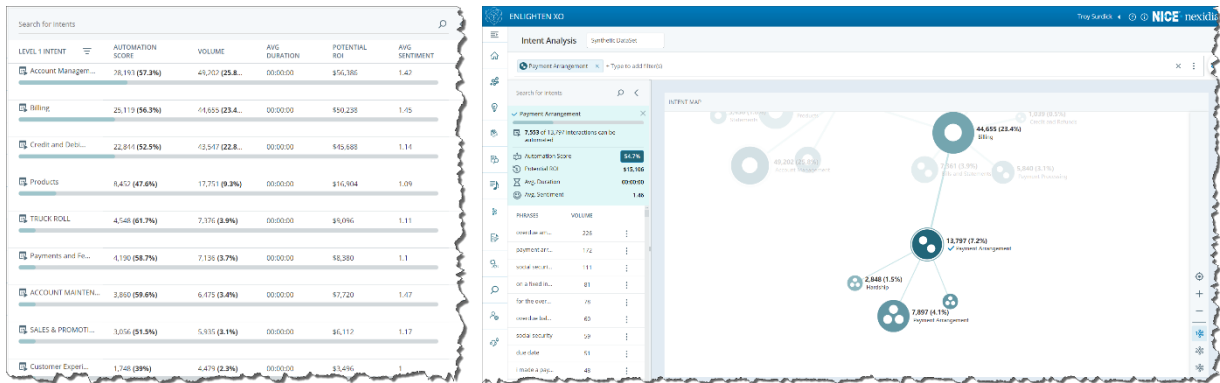
Step 3: Investigate Customer Intents

Intent Analysis helps the user understand which interactions are the best candidates for automation efforts. The system provides the user with relevant data about each Customer Intent (that is, why the end customer contacted the enterprise). For any given Intent, the user sees volume per interaction type, Automation score, Potential ROI (privileges required), Average Sentiment, and Average Duration. Based on this information, the user can quickly select intent/s that s/he would like to work on.

Customer Intents are available to be viewed in a spatial (network diagram) or an expandable tabular layout (see below), allowing the user to easily navigate the intent hierarchy, or to sort and compare different Intents across the metrics noted above. Selecting an Intent within the user interface provides detailed data about that intent, as well as key phrases associated with it, allowing the user to easily access the actual agent-involved interactions that were the basis for the analysis, and starting from v12.4 FP5.2 X4, downloading the list of the relevant interactions for each Intent.

Note – as of v12.4 FP4 (March 2023), the node images have been updated based on user feedback from what is pictured below to a slightly different paradigm. Rather than having Level 1 nodes being fully enclosed circles as indicated in the screen shots in this document, the images have been updated such that Level 1 nodes have one inner circle, Level 2 nodes have two inner circles, and Level 3 nodes have three inner circles, as pictured here: .





Filter Intents

Users can simultaneously filter customer Intents by these metrics (orange rectangle pictured at right): Volume, Automation Score, Number of Tasks, Potential ROI, Average Duration, and Sentiment. Starting with v12.4 FP5.2 X6, the system can allow direct numeric text entry into fields in conjunction with sliders. Thus, the user can quickly and easily identify customer Intents that are the best candidates to spend their time working on.



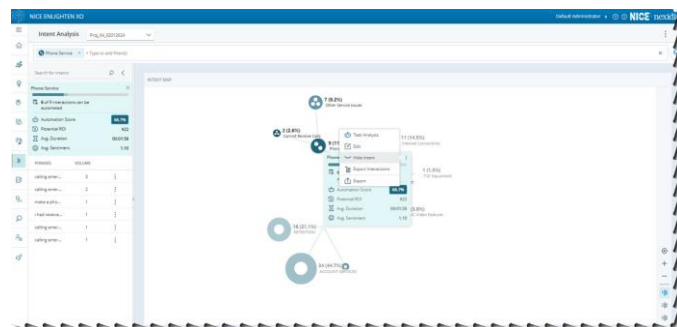
Users can also focus on specific interactions within the Project via the Automatable and Media Type filters (blue rectangle).

Hide Intents

Starting with v12.4 FP5.2 X5, users can hide customer Intents starting from every level by clicking on the “Hide Intent” button from the context menu.

This allows the user to temporarily concentrate on relevant intents, ignoring intents that might not be relevant to the business.

This action automatically hides all its sub-intents; in case the hidden intent is Level 1 or Level 2. When hiding an intent, all other Percentage metrics are re-calculated.



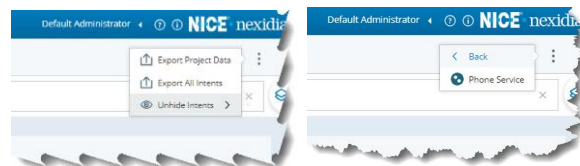
without the hidden intent's data within the spatial (network diagram) and the expandable tabular layout.

The hidden intents do not appear on the following views:

- Intents Dashboard
- Data Exports
- Task Analysis Filters
- Intents Trending Report
- Any calculation related to the overall intent data



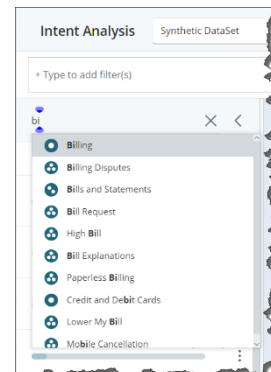
To unhide intents, click the 'Unhide Intents' button (shown on the right) on the Intent Analysis and choose an intent, or as part of the XO Objects on the Project Management page.



Search for Intents and Trend Intents

The Intent Analysis page supports searching (across levels) for Intents.

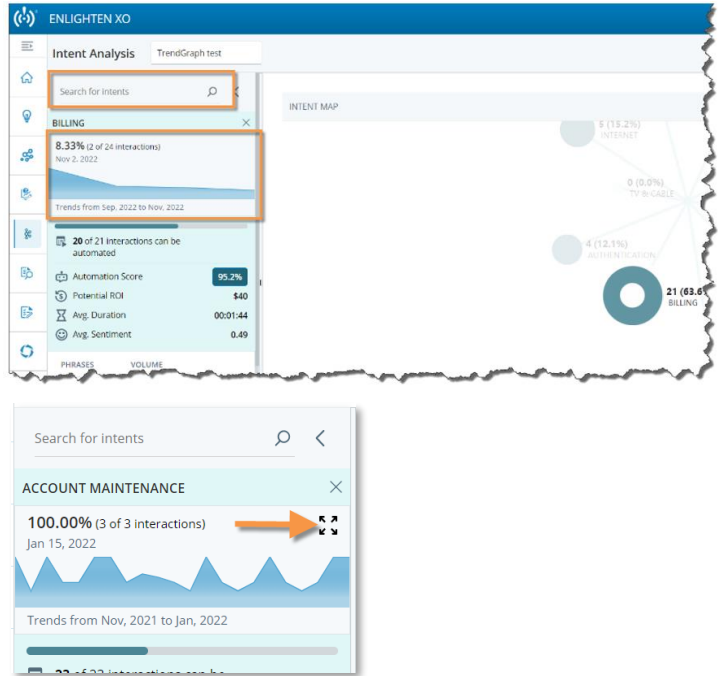
Intent Searching uses a “contains” method, so that any intent whose name includes the entered characters – regardless of their position will appear in the results. This is illustrated in the example of “Mobile Cancellation” shown on the right.



When a project is configured for Trending, the system will utilize idle time to compute and store the primary intent for interactions that come into the Target Media Set used for that Enlighten XO project after the project is initiated. This data is used for the trend chart pictured here, showing volume for the intent from the date that the project was run.

Additional data / enhanced trending dashboard is available for view via the “expand” widget pictured here. For more about that dashboard, see:

Enhanced Intent Trending, later in this document.



Step 4: Agent Tasks and Task Flows

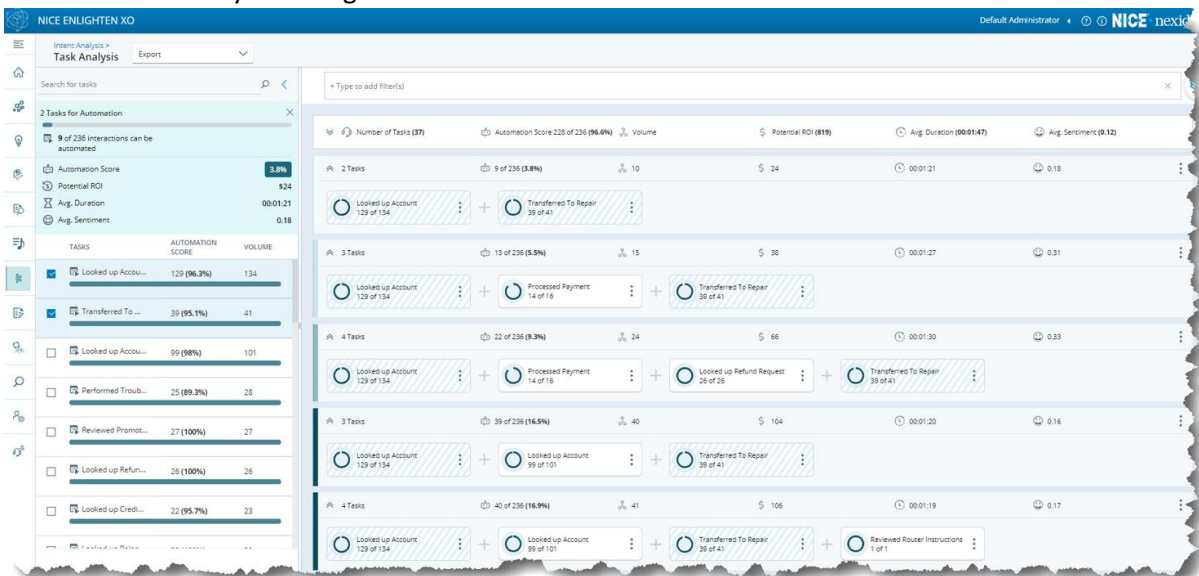
Task Analysis / Task Sets

Once the user has identified a specific Intent to dig into, the Enlighten XO system provides data about the individual agent Tasks that were identified to be present in the Interactions with this selected Intent. The system provides groups of Agent Tasks that occur together in these specific interactions (Interactions with the selected Customer Intent), as well as data to indicate the potential benefit / containment should those tasks be automated.

Both the number of these Task Groups (default = 30) and the number of Task Groups containing 2, 3, 4, etc. individual Agent Tasks (default = 2 each of 2-task groups, 3-task groups, etc., depending upon what occurs in the actual interactions) are configurable within the system. Should the user have specific tasks that they wish to include and/or have already worked on or completed, they can update their view to return Task Groups containing those tasks. Starting from v12.4 FP5.2 X5, user can search task on the Task List (left panel) using the search icon and typing characters (up to max length of 25) in the search field. This will result in a drop down of any task containing the entered string, with the ability to choose one or more tasks. The chosen tasks in the drop down will be reflected as checked in the Task List.



In the example below, the user has indicated that “Looked up Account” and “Transferred To Repair” must be included by selecting them in the Task List.



- All Task Groups presented contain the selected Tasks (potentially including others), and they appear as “highlighted” in their groups.
- Summary area provides data to user about what they may accomplish / contain by automating the selected tasks (e.g., 17 of the 860 interactions could potentially be automated if the Agent Tasks “Looked up Account” and “Transferred To Repair” were both automated).
- Note: as of v12.4 FP4.1.2, individual tasks in the left-side list have checkboxes to facilitate selection / de-selection.

- Note, as of v12.4 FP 5.1 X3 (December 2023) there is a new Volume column on this page that enables users to easily understand the total volume of interactions (not just the count of automatable interactions) within their current context (e.g., Intent) that contains each set of tasks.

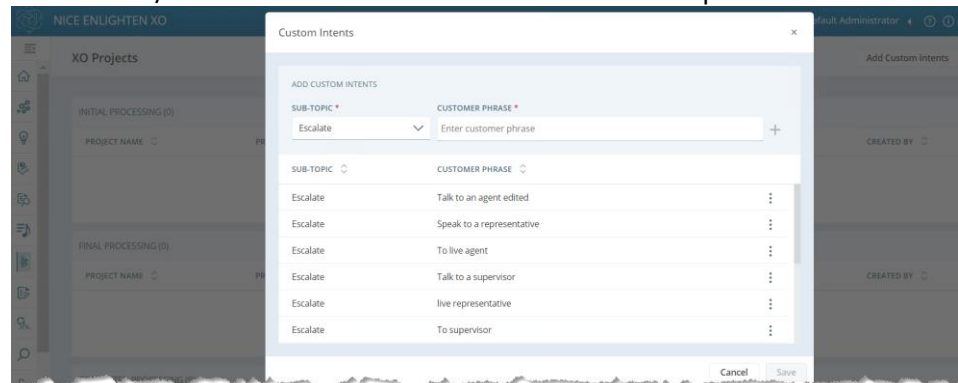
Data for each suggested Group of Tasks includes number of unique Agent Tasks, Automation Opportunity (that is, how many of the interactions in the selected Intent contain only the Agent Tasks presented in this group), Potential ROI (visibility controlled by privilege), Average Duration, and Average Sentiment. Thus, at this stage the user can focus on the specific Agent Tasks to work on within their selected Customer Intent.

Note: On the Task Analysis screen, the Interaction Count presented for each Task Set/Group includes interactions containing ALL or any SUBSET of the tasks included in that Task Set.

Step 5: Add Default Custom Intent Phrases

Starting with v12.4 FP5.2 X5, the users will have the option to select the standard Intents and their associated phrases as part of the export files. Standard intents are tailored to cater to distinct business requirements. By default, a custom intent focused on Escalation and its associated customer phrases is included. This intent can be exported from the Export Intent page. Additional custom intents can be added based on the specific needs of the business.

Below is the example where the user can add/edit the Custom Intents and their associated phrases:

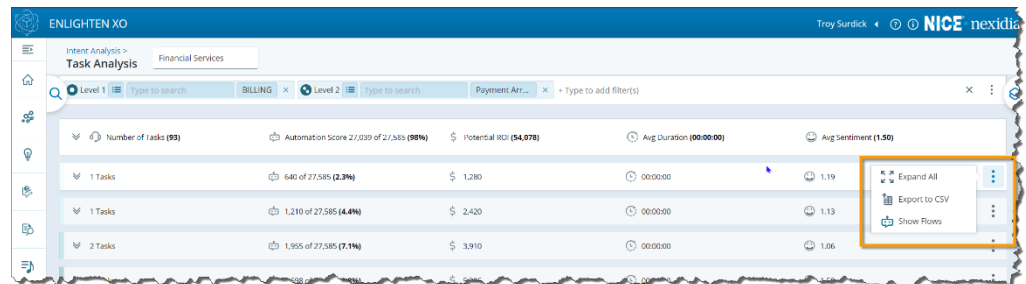


Only users with the Enlighten XO – Project Administration privilege can add custom intents. The custom intents added will be available on the Export Intent page.

INCLUDE	CATEGORY	TOPIC	SUB-TOPIC	CUSTOMER PHRASES	VOLUME (N)	AUTOMATION SCORE	AVG DURATION	AVG SENTIMENT
☒	PHONE	Phone Service	Other Service Issues	calling america phone ser...	1 (20%)	100%	00:01:23	0.03
☒	BILLING	Payment Method	Pay By Bank Account	the bank account on your...	1 (100%)	100%	00:00:54	0.81
☒	ACCOUNT MAINTENANCE	Account Management	Update Mail Address	mailing address	1 (100%)	100%	00:03:22	0.46
☒	CUSTOM	custom	Escalate	Talk to an agent edited	-	-	-	-
☒	CUSTOM	custom	Escalate	Speak to a representative	-	-	-	-
☒	CUSTOM	custom	Escalate	To live agent	-	-	-	-
☒	CUSTOM	custom	Escalate	Talk to a supervisor	-	-	-	-
☒	CUSTOM	custom	Escalate	live representative	-	-	-	-
☒	CUSTOM	custom	Escalate	To supervisor	-	-	-	-

Task Flows

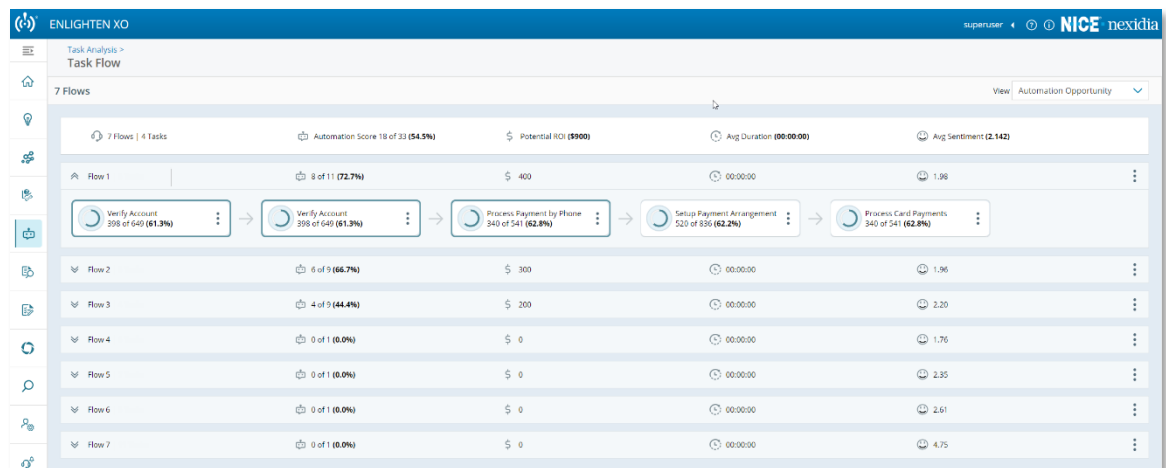
Once the user has identified the set of Agent Tasks to focus on, s/he can view Task Flows – the top “flows” (occurrence) of how those tasks appeared in the interactions.



Users can easily see which flow/ordering had the highest volume, sentiment, automation opportunity, and more to use as a guide for structuring the flow of the self-service engine.

In the example below, the user has selected a group containing 4 individual Agent Tasks (where a given task may repeat within a flow) and is viewing the Task Flows based on Automation Opportunity. We can see that the top flow in this case is: Verify Account → Verify Account → Process Payment by Phone → Setup Payment Arrangement → Process Card Payments.

Note: the interaction counts in the Task Flow screen represent the interactions that contain ALL of the tasks from the selected task set (four, in this example), exhibiting the exact selected flow.



Users can also export Transcripts (up to 10) representative of a selected Task Flow, as of the v12.4 FP4.2 release (May 2023).

The screenshot shows the 'Task Flow' section of the ENLIGHTEN XO interface. It displays a table with 14 flows. The table has columns for flow name, automation score, potential ROI, average duration, and average sentiment. A context menu is open for 'Flow 1', showing options like 'Expand All', 'Export Transcripts', 'Export to AutoFlow', and 'Export to CSV'.

Flow	Automation Score	Potential ROI	Avg Duration	Avg Sentiment
14 flows	Automation Score 11 of 21 (52.4%)	\$ Potential ROI (\$22)	00:00:00	1.626
Flow 1	3 of 4 (75.0%)	\$ 6	00:00:00	1.23
Flow 2	3 of 3 (100.0%)	\$ 6	00:00:00	2.66
Flow 3	0 of 3 (0.0%)	\$ 0	00:00:00	1.00
Flow 4	1 of 1 (100.0%)	\$ 2	00:00:00	1.56
Flow 5	1 of 1 (100.0%)	\$ 2	00:00:00	2.31

Ancillary / Supporting Functionality

Object Management

Enlighten XO allows users (with appropriate Privilege) to perform various operations on Intents, Tasks, and Phrases within a Project to track and see their progress. Intents and Tasks can be renamed, hid and can be marked (or un-marked) as having their Work Complete.

Phrases can be hidden or un-hidden.

The screenshot shows the 'XO Objects' section of the NICE ENLIGHTEN XO interface. It displays a table with columns for object name, type, hidden status, status, and original label. The table lists several objects, including 'TECH SUPPORT', 'Retention Changes', 'I need to pay my wireless bill', 'Discontinued Credit Request', 'ACCOUNT ONLY', 'credit your account for the month', and 'Phone Service'.

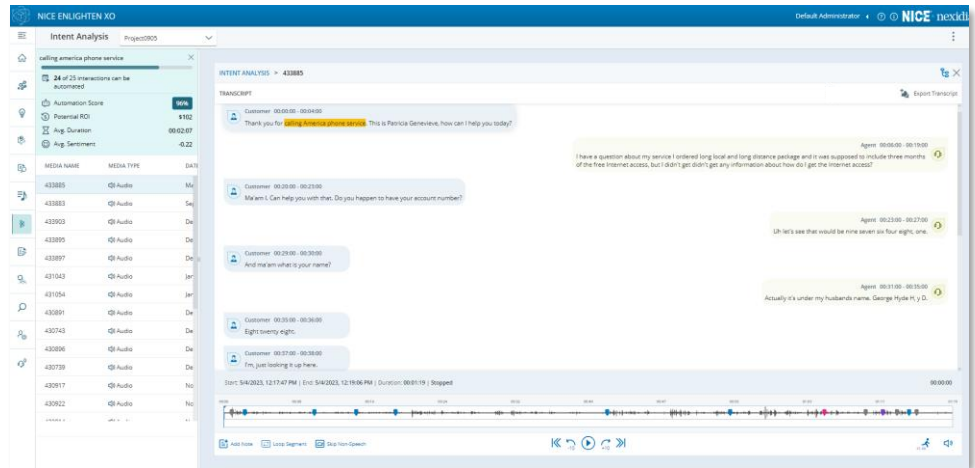
OBJECT (S)	TYPE	HIDDEN	STATUS	ORIGINAL LABEL
TECH SUPPORT	Intent L1		✓ Work Complete	
Retention Changes	Intent L2			Retention Service Changes
I need to pay my wireless bill	Phrase	✓ Hidden		
Discontinued Credit Request	Task		✓ Work Complete	
ACCOUNT ONLY	Intent L1			ACCOUNT SERVICES
credit your account for the month	Phrase	✓ Hidden		
Phone Service	Intent L2	✓ Hidden		

Starting with v12.4 FP5.2 X5, intents can be hidden or un-hidden.

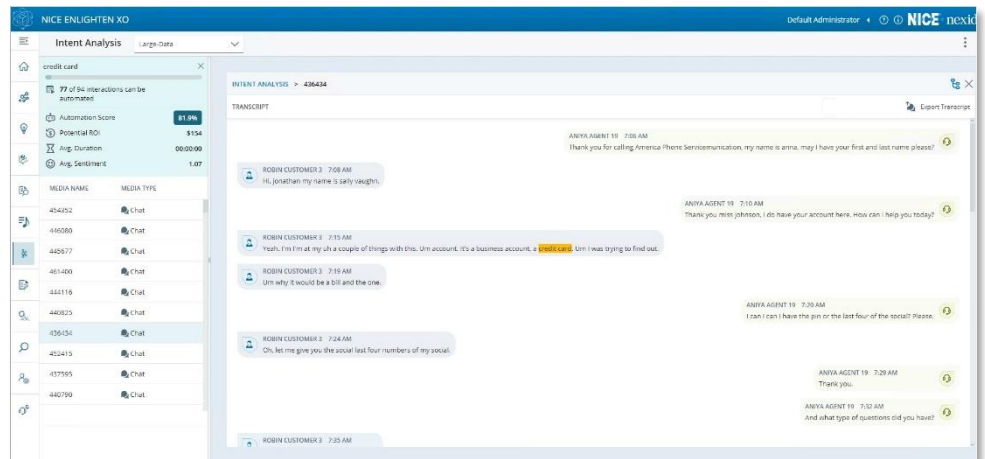
Users (again, assuming appropriate Privilege) can access all the modified objects via a separate object management administration page (Accessed from the Enlighten XO Projects page). Such users can restore objects to their original status, if desired.

Playback and Transcripts

In various places on the user interface, users can View Interactions (e.g., interactions containing a given Phrase). Customers with Enlighten XO and Interaction Analytics can additionally play back the audio file (call), as pictured here:



For playback of Chat interactions (or in the case of Enlighten XO standalone deployment), the full transcript remains visible, but without the audio playback controls.



As of v12.4 FP5.2 X5, users can export the specific interaction transcript in text file format by clicking on the 'Export Transcript' button in the upper right corner of the transcript viewer area.

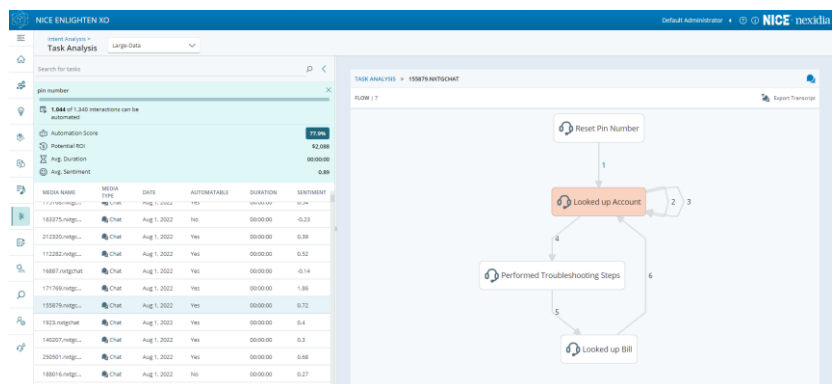
As of v12.4 FP5.2 X6, the user will select an interaction from a table grid, as seen in the screenshot above.

Expanding the left pane will provide the following metrics in the grid columns: media name (to find it on different applications such as search), media type, its date, if it's automatable, its duration, and the sentiment score.

MEDIA NAME	MEDIA TYPE	DATE	AUTOMATABLE	DURATION	SENTIMENT
415813	Audio	May 1, 2018	Yes	00:00:48	-0.88
415791	Audio	May 1, 2018	Yes	00:01:03	-0.88
415801	Audio	May 1, 2018	Yes	00:01:01	-0.28
415587	Audio	May 1, 2018	Yes	00:01:12	-0.88
415776	Audio	Apr 1, 2018	Yes	00:00:58	-0.46
415583	Audio	Apr 1, 2018	Yes	00:01:17	-0.46
415578	Audio	Mar 1, 2018	Yes	00:00:58	0.05

Individual Interaction Task Flow

Users can also view individual tasks that occurred in the selected interaction. The system provides data about the order of the tasks via numbering and highlighting/animation.



Data Export

The user can export the data of any level of Intent from the Intent Analysis “Dashboard” to CSV. It will export all the information about all the metrics that are shown in the Enlighten XO UI.

As of v12.4 FP5 (July 2023), the application will identify six different types of entities, Address, Currency, E-mail address, Location, Name and Number, and include those in Exports. So, for example, rather than the export containing “one two three” it would contain “[NUMBER] [NUMBER]”.

As of X8 (September 2024), Enlighten XO supports exporting (via AutoFlow) intent data and conversational flow data formatted for multiple bot building applications as described in [Breaking Changes](#), above.

Limitation: This initial version of Entity detection is more effective for audio interactions than chats. For example, “me at mail dot com” as transcribed from a call is more likely to be identified as [EMAIL] than

“me@mail.com”, which is likely how that would be input into a chat. Subsequent releases will continue to enhance the entity detection capability.

Export Intents:

The user can configure and preview the customer phrases before exporting them in a generic CSV format or AutoFlow format that is formatted specifically for the consuming bot application.

- The user can (optionally) have the system group phrases into phrase groups (clustering phrases that are similar to each other), maximum up to 20.
- The user can configure however many, maximum up to 9999, customer phrases to export.
- The user can unmask/mask data for previewing and before exporting them. Masking shows entities as pictured below, rather than actual raw data. Unmasking shows the actual raw data.
- As of v12.4 FP5.1 X1:
 - We have a word count filter, enabling users to filter the Customer Phrases according to their length.
 - The Export to CSV functionality will always contain all three levels of Intent (Category, Topic, Sub-Topic) to facilitate user understanding that full hierarchy
- As of v12.4 FP5.1 X3 new filters include:
 - Volume (%) filter, enabling user to filter phrases by the percentage of interactions in <the Intent> that contain that phrase. Note that as part of this enhancement, the Volume column in the Intent Export is updated to show the number of interactions in <this Intent> that contain at least one hit for <the phrase>. (Previously it showed the number of times the phrase was found, which could have resulted in a given interaction being counted more than once.)
 - Automation Score filter, enabling user to filter by the Automation Score (average) for those interactions in <this Intent> containing that phrase.
 - Phrase Format – allows users to choose between the ‘full’ phrase and a shorter system-generated snippet.
 - These pre-existing filters have been moved into a collapsible Filters panel, which is expanded and collapsed via the Filter icon .
- As of v12.4 FP5.2 X4 a new filter includes:
 - Free Text Phrase Filter - users to filter phrases by specific terms/words, allowing for a detailed search of phrases containing those terms to be utilized in the bot builder
- As of v12.4 FP5.2 X8, the “Customer phrases per sub-topic” filter allows the ability to choose from several operators, as seen below. The default filter will show all phrases between 5 and 50.

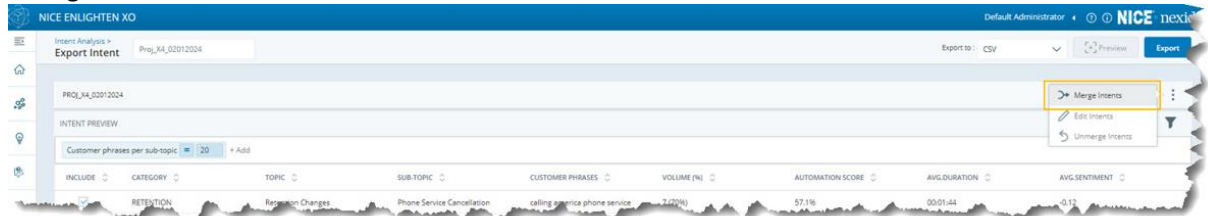
INCLUDE	CATEGORY	SUB-TOPIC	CUSTOMER PHRASES	VOLUME (%)	AUTOMATION SCORE	AVG. DURATION	AVG. SENTIMENT
☒	ACCOUNT MAINTENANCE	Account Management	Update Account Profile	544 (32.7%)	79%	-	0.9
☒	ACCOUNT MAINTENANCE	Account Management	Update Account Profile	94 (5.7%)	90.4%	-	1.19
☒	ACCOUNT MAINTENANCE	Account Management	Update Account Profile	8 (0.5%)	100%	-	1.48
☒	ACCOUNT MAINTENANCE	Account Management	Update Account Profile	9 (0.5%)	55.6%	-	1.31
☒	ACCOUNT MAINTENANCE	Account Management	Update Account Profile	7 (0.4%)	71.4%	-	0.59
☒	ACCOUNT MAINTENANCE	Account Management	Update Account Profile	3 (0.2%)	100%	-	0.75
☒	ACCOUNT MAINTENANCE	Account Management	Update Account Profile	3 (0.2%)	100%	-	1.36
☒	ACCOUNT MAINTENANCE	Account Management	Update Account Profile	3 (0.2%)	100%	-	1.37
☒	ACCOUNT MAINTENANCE	Account Management	Update Account Profile	3 (0.2%)	100%	-	0.64

- As of v12.5 FP1 X10, the “Recommended Phrases” option will allow users to export only the best, most relevant and balanced set of customer phrases for each intent. The system uses a new algorithm to select phrases based on the following:
 - Relevance to the intent (semantic distance)
 - Diversity across label groups
 - Balanced representation across intents
 - Phrase quality based on volume and word count)

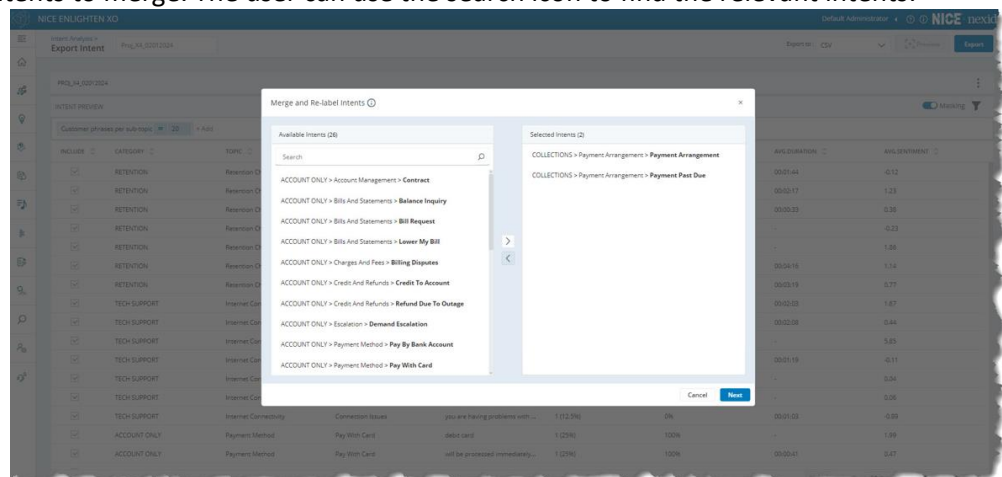
When exporting Intents using the “AutoFlow” format, the data grid in the User Interface (as of v12.4 FP5.1 X3, December 2023) will contain columns for Automation Score and Volume. Those columns are ONLY in the user interface and are for the purpose of confirming that the corresponding filters behaved as expected. The resulting TSV file format has not changed and still contains only the Intent and the Phrase/Utterance.

INCLUDE	SUB-TOPIC	CUSTOMER PHRASES	VOLUME (%)	AUTOMATION SCORE
☒	Phone Service Cancellation	calling america phone service	7 (70%)	57.1%
☒	Phone Service Cancellation	from your home phone	4 (40%)	75%
☒	Phone Service Cancellation	calling america phone services	1 (10%)	100%
☒	Phone Service Cancellation	i am home phone	1 (10%)	100%
☒	Phone Service Cancellation	i have got a phone call on the phone service	1 (10%)	100%
☒	Phone Service Cancellation	i make a phone service	1 (10%)	0%
☒	Phone Service Cancellation	to stay with them like a phone service	1 (10%)	0%
☒	Connection Issues	am you connected internet access	2 (20%)	100%

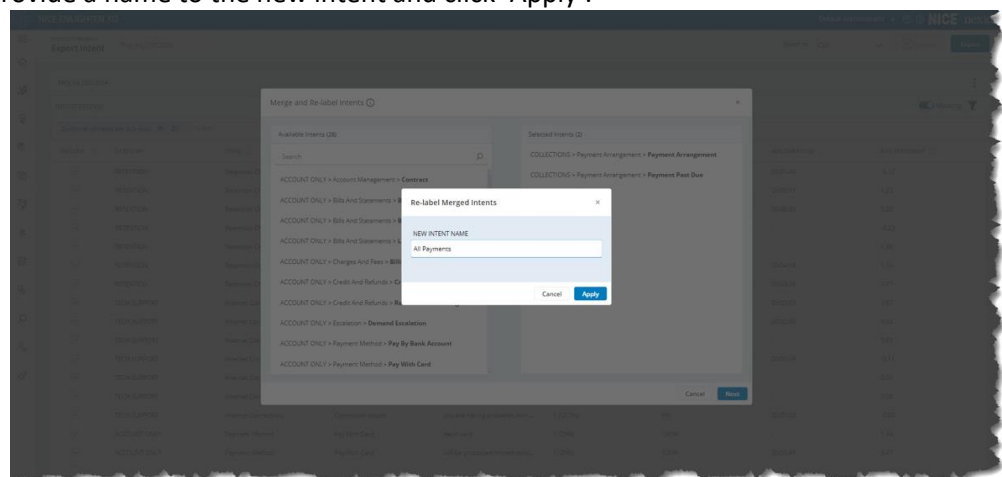
As of v12.4 FP5.2 X5 user can temporarily create a maximum of 10 merged Level 3 intents for a given export, each with as many Level 3 intents as needed by clicking on the below item (orange rectangle, below). For cases where multiple intents need to follow a common flow, the merging of intents makes it easier to artificially / arbitrarily group those intents. A common example would be the case where automation efforts are focused on Topic A, so (for now, at least) any interaction that is determined to be about Topic B “any of <these intents> should go directly to a live agent, or first authenticate and then go to live agent.



This will bring the user to a ‘Merge and Re-label Intents’ page (pictured below) where they can choose any number of L3 intents to merge. The user can use the search icon to find the relevant intents:



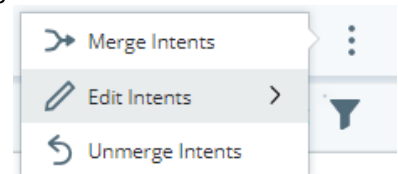
The user needs to provide a name to the new intent and click ‘Apply’:



The new merged intent will be reflected in the export phrase list below. Each phrase from the original merged phrase will be part of the new merged intent:

INCLUDE	CATEGORY	TOPIC	SUB-TOPIC	CUSTOMER PHRASES	VOLUME (N)	AUTOMATION SCORE	AVG DURATION	AVG SENTIMENT
☒	RETENTION	Retention Changes	Account Cancellation	I am calling to cancel it	1 (33.3%)	100%	-	4.97
☒	RETENTION	Retention Changes	Account Cancellation	why do you want to close your acc...	1 (33.3%)	100%	00:02:53	0.54
☒	RETENTION	Retention Changes	Account Cancellation	why you and to close your acc...	1 (33.3%)	0%	00:01:39	1.82
☒	COLLECTIONS	Payment Arrangement	All Payments	I made a payment	1 (100%)	100%	-	2.58
☒	COLLECTIONS	Payment Arrangement	All Payments	overdue amount we are ha...	1 (100%)	100%	-	2.58
☒	COLLECTIONS	Payment Arrangement	All Payments	payment arrangement has be...	1 (100%)	100%	-	0.79
☒	COLLECTIONS	Payment Arrangement	All Payments	payment arrangement has be...	1 (100%)	100%	-	0.97

The user can edit a merged intent or to un-merged an intent, by clicking on the relevant button:



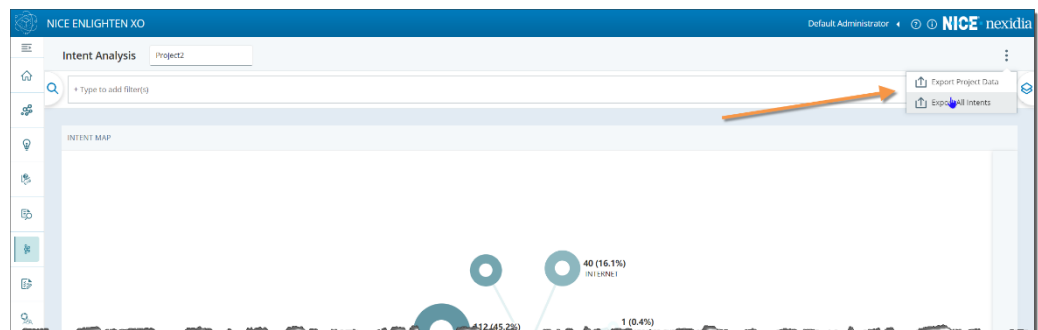
User can merge, edit merged intents and un-merge intents from both “CSV” and “AutoFlow” export formats.

Note: The intents will be merged temporarily only for the specific export process. They are not saved anywhere else and are not affecting any other page within Enlighten XO.

Export Enlighten XO Project Data:

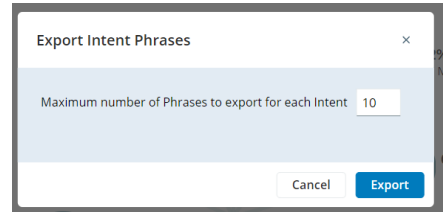
As of v12.4 FP5.1 (September 2023), users with Export privileges have the option to initiate a “whole project” export from the Intent Analysis page.

As of v12.4 FP5.1 X1 (October 2023), users with Export privileges also have the option to initiate an “all intents” export from the Intent Analysis page. This allows users to export data for all Intents within the XO project at once.



This Export Project Data option produces a JSON file that contains data about each Level 3 Intent within the XO Project, as well as a selected Task Flow for each.

- Intent Phrases – will include up to a user-entered number of Intent Phrases for each intent. User is presented with the dialog box shown below upon selection of the Export Project Data option:



Export Intent Phrases

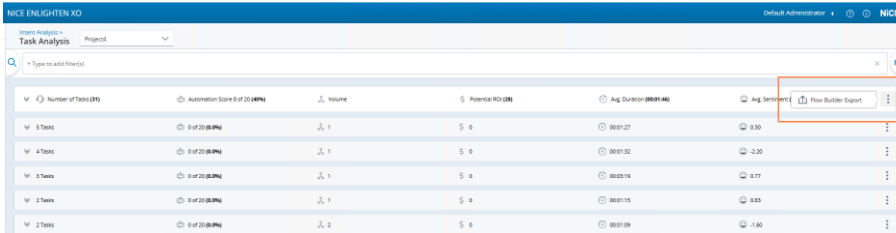
Maximum number of Phrases to export for each Intent 10

Cancel Export

- Task Flow – selected based on these criteria. The selected task flow is determined by a rule sequence, which are:
 - Least number of tasks steps, with a minimum of two tasks.
 - If the above results in multiple flows, choose the Task Flow with the most automatable interactions.
 - If the above results in multiple flows, choose the Task Flow with the highest average sentiment score.
 - Once the Task Flow has been determined, if there is more than one interaction containing <that flow>, choose the interaction with the highest sentiment score.

Export Interactions – Task Analysis Page:

Note, as of v12.5 FP 1 X10, The Task Analysis page allows you to export interaction data for all task sets. This data can be used by external tools in the Automation Flow Builder. Users can choose to export data using the header options for the task sets currently visible in the Task Set grid.



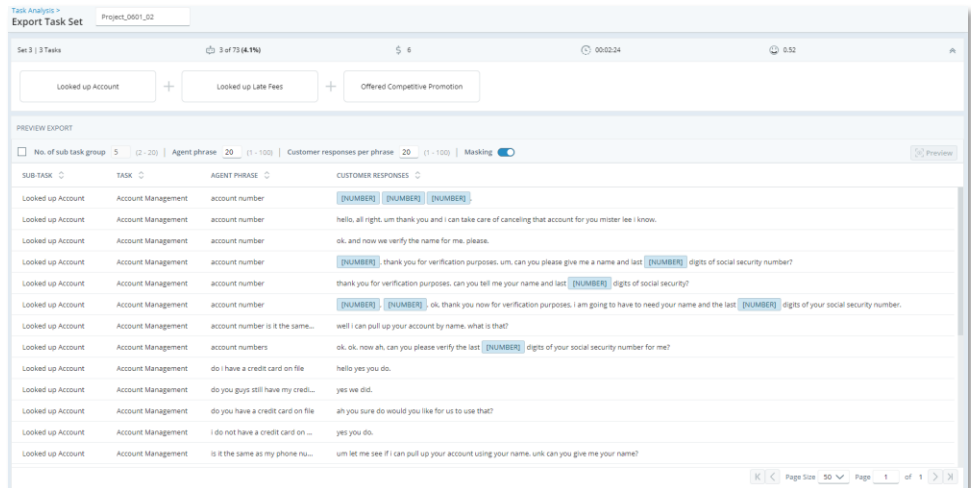
The screenshot shows the 'Task Analysis' page with a table of task sets. A red box highlights the 'Flow Builder Export' button in the header of the table.

Number of Tasks (15)	Automation Score (0 of 20 (40%))	Volume	Potential ROI (\$)	Avg. Duration (00:01:46)	Avg. Sentiment	Flow Builder Export
5 Tasks	0 of 20 (0%)	1	\$ 0	00:01:27	0.90	
4 Tasks	0 of 20 (0%)	1	\$ 0	00:01:02	0.20	
3 Tasks	0 of 20 (0%)	1	\$ 0	00:02:19	0.77	
2 Tasks	0 of 20 (0%)	1	\$ 0	00:01:15	0.83	
2 Tasks	0 of 20 (0%)	2	\$ 0	00:01:09	0.80	

Export Individual Tasks or Task Sets:

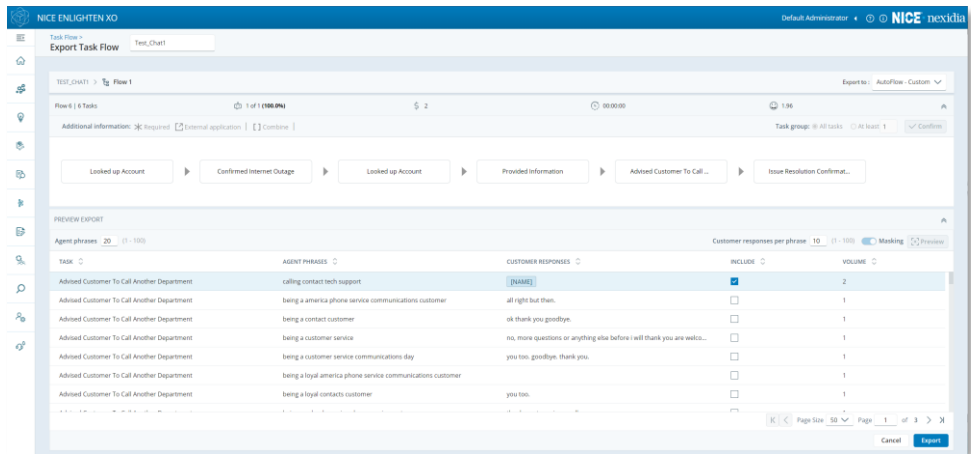
The user can configure and preview the agent phrases and customer responses before exporting them in CSV format.

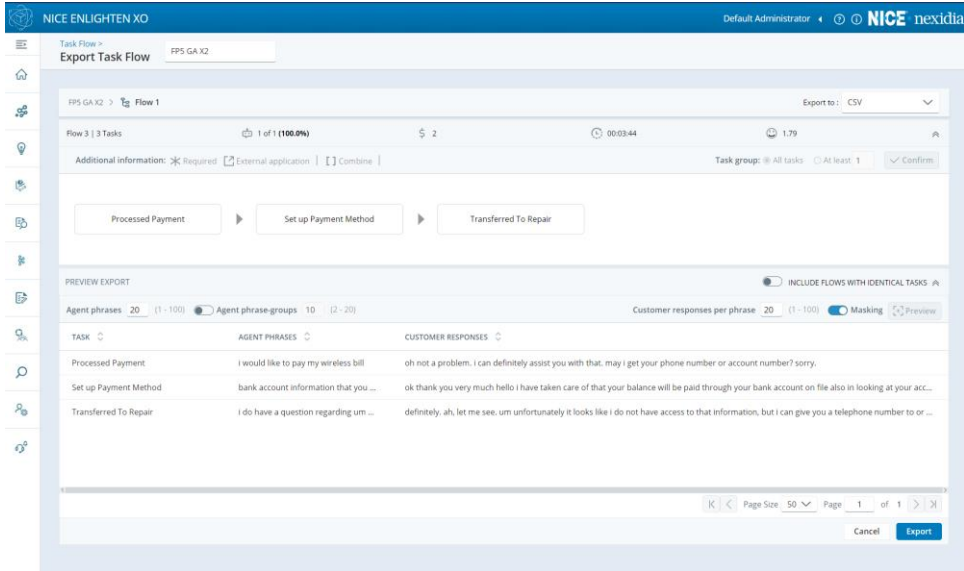
- The user can have the system group phrases into phrase groups (the phrases that are very similar to each other), maximum up to 20.
- The user can configure however many, maximum up to 100, agent phrases that s/he would like to export.
- The user can configure however many, maximum up to 100, customer responses per agent phrase that s/he would like to export.
- The user can unmask/mask data for previewing and before exporting them.
- The user can preview the data after configuring and before exporting it.



Export Task Flow

As of v12.4 FP5.2 X8, Export to BPMN is no longer available.

Export Format	Details / Notes
“AutoFlow - Example” (JSON)	As of v12.4 FP5.2 X8, this option is available only to customers with a specific bot / SmartAssist integration configuration. The system will select an interaction that matches <this flow> and export that conversation into JSON format.
“AutoFlow – Custom” (JSON)	As of v12.4 FP5.2 X8, this option is available only to customers with a specific bot / SmartAssist integration configuration. New option in v12.4 FP5.1 (September 2023) results in two JSON files being exported: Entities.JSON, and Conversation.JSON. User can configure the Tasks / ordering as they do in CSV, including (optionally) marking a given task as likely needing (interface with) External Application.  Configure task order

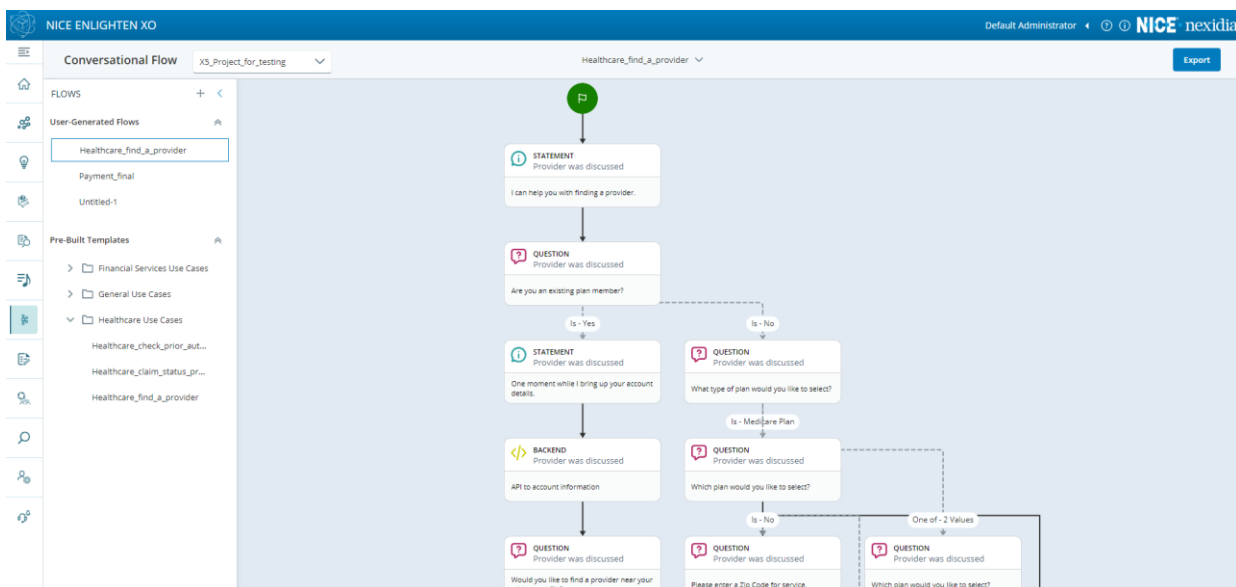
Export Format	Details / Notes
	- For each task, select which Agent Phrase / Customer Response combinations to include in the “conversation” to be exported. - Default = the most common / highest volume combination per task. If more than one combination is selected, the highest volume will be included in the JSON conversation export. - Phrases / Responses are included in the order matching task configuration from Enlighten XO export page. - When customer response is a named Entity – the bot will include that Entity as the positive response, and default the negative response to <not that Entity>, and default the “I don’t know” case to the standard IDK option in bot. - If any task is marked as “External Application”, a “Backend” placeholder will appear in the Digital Employee Builder at the corresponding location in the conversation, reminding the bot builder to establish that external application interface. - Conversation.JSON file includes the above content and can be imported directly into the bot’s “Digital Employee Builder”. Entities.JSON includes all entities from Customer Responses selected to be included in the export, streamlining Entity creation within bot.
CSV	The user can configure tasks from the selected Task Flow as desired before exporting task flow data:  - Each task can be marked as required or not. - The task can be marked whether it requires External Application lookup to be completed (e.g., validating a customer’s account number or address). - Multiple tasks within the flow can be combined into one step. - Tasks can be moved from left to right and vice versa as needed to indicate the sequence/ordering of the tasks. - If there are multiple tasks in a group, it can be marked whether all those tasks are needed vs. a subset. - Agent Phrases and Customer Response data can be configured for CSV export as it is for Tasks and Task sets.

Export Format	Details / Notes
	- The “Include flows with identical tasks” toggle allows users to view phrases and responses from interactions that contain the same tasks selected for this export, but in a different order. This provides the user with additional real conversation/interaction data to select from. - The tasks configuration and phrase data are exported in two CSV files (one for the task configuration data, another for the Agent Phrase / Customer Response data per Task)

Conversational flow

The new conversational flow allows end users to configure a custom experience by utilizing arbitrarily selected tasks and conversational utterances from XO. This enables personalized interactions, providing more flexibility and control over the sequence and content.

This functionality, added in v12.4 FP5.2 X8, results in a compatible file that can be imported to the bot.



- The following are the types of actions that can be selected as part of the flow design:
 - Question – An action type that defines the bot to ask a question and expect a response. The user needs to select a task, and as a result, the list of agent and customer response snippets from that task will be chosen for the bot to present. As of X8, the user can edit an existing phrase or add a phrase ad-hoc. The user defines the variable name that stores the answer, and the variable type is Free-Text.

Starting from version 12.5 X9, for specific supported bots, the user can choose seven more variable types: Number, Alpha, AlphaNumeric, Date, Percentage, Currency, Boolean

▼ VARIABLE TYPE

Alpha
AlphaNumeric
Boolean
Currency
Date
Free Text
Numeric
Percentage

- Starting from version 12.5 X9, certain supported bots can now utilize multiple agent phrases for a question action type. This enhancement provides bots with the flexibility to rephrase questions when needed. If a response is not received or an error occurs, the bot can present alternative phrases defined by the user to retry the question.

Properties >

> ACTION TYPE

QUESTION

> AGENT TASK

Looked up Account

QUESTION	RESPONSE
> AGENT PHRASE	
it is just the same as my phone number	
FIRST ALTERNATIVE PHRASE	
that number is it the same as my phone	
> SECOND ALTERNATIVE PHRASE	
what is the account number	

- Statement - An action type that defines the bot to present a sentence without expecting to get any response. The user needs to select a task, and as a result, the list of agent snippets from that task will be chosen for the bot to present.
- Backend - An action type where the bot calls an external application (like an API) to retrieve or verify data. For example, the API calls a CRM that holds customer data and sends the last 4 digits of SSN and last name to verify identity. This action type serves as a placeholder, and the exact external application needs to be defined and configured in the bot building tool. As of version 12.5 X9, the Backend Block now includes two additional fields: "Input Variable" and "Output Variable".
The Input Variable dropdown contains all previously defined variables for any action type (Question, Output Sub-Flow, Output Backend). This optional field allows the end-user to select one variable if needed.

The Output Variable field is free text, enabling users to create a custom variable, and is paired with a dropdown called "Output Variable Type", similar to the Question action type but with properties specific to the Backend.

- Escalation - This action type signals the bot to escalate this flow further to an agent.
- Sub Flow - An action type that serves as a placeholder in the chatbot that triggers a **sub-flow** or different flow, acting as a block within the conversational flow.

Starting from version 12.5 X9, the Sub Flow Block now includes two additional fields: "Input Variable" and "Output Variable," with functionality identical to that described for the Backend Block. Starting with version 12.5 FP1 X10, a new "Link Subflow" option has been introduced, allowing users to select and reuse subflows from a drop-down menu in other flows. This feature is available only for users with the configuration parameter **Nexidia.XO.BOT** set to **Omilia**.

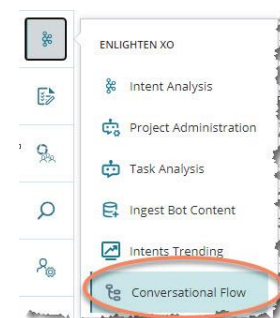
- End Flow - This action informs the bot that the flow is complete, indicating that the user's needs have been met.
 - Anything Else - Starting from version 12.5 FP1 X10, added "Anything Else" action block specifically designed for Omilia-based flows to let bots ask users if they have additional intents.
- As of 12.5 X9, for specific supported bots, the user can define data privacy for the Question, Statement, and Backend. Depending on the sensitivity of the information the bot's user provides, the conversational flow builder can decide whether this data will be protected. If you want to protect the user data, you can choose how it will be stored.

The three options are

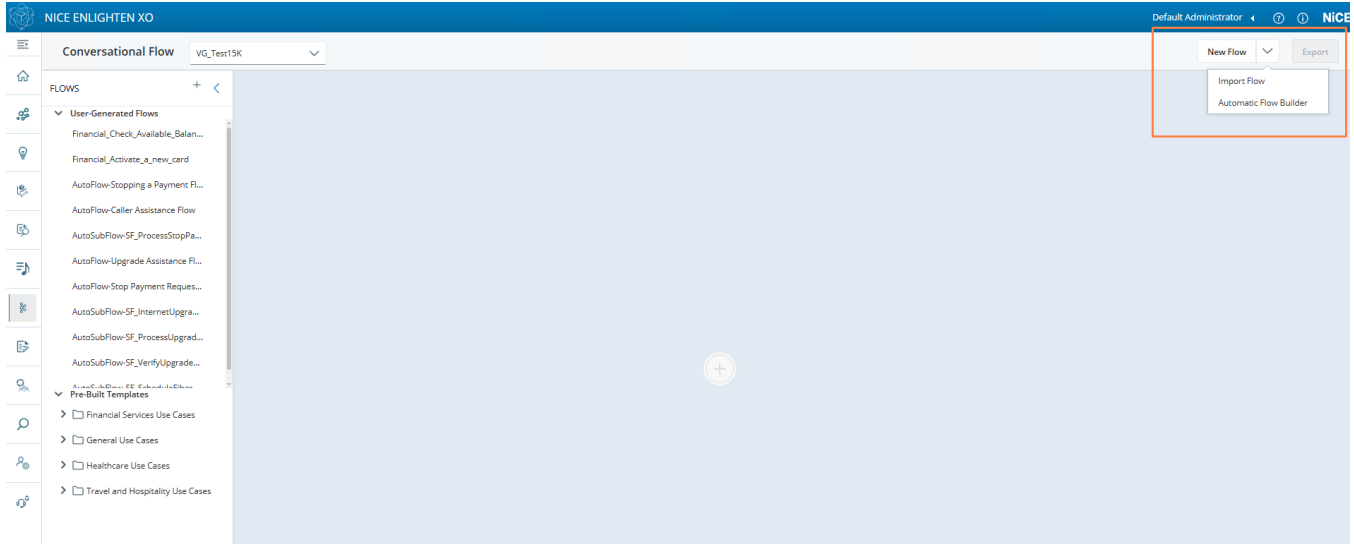
- Masked Data – Bot stores the data as masked text.
- Encryption – Bot stores the data as encrypted.
- None - stores the data as plain text

- The user can navigate to the conversational flow from the following area:
 - XO Main menu:

- In this scenario, once the user clicks "Conversational Flow" option, user will be directed to the Conversational Flow Page.

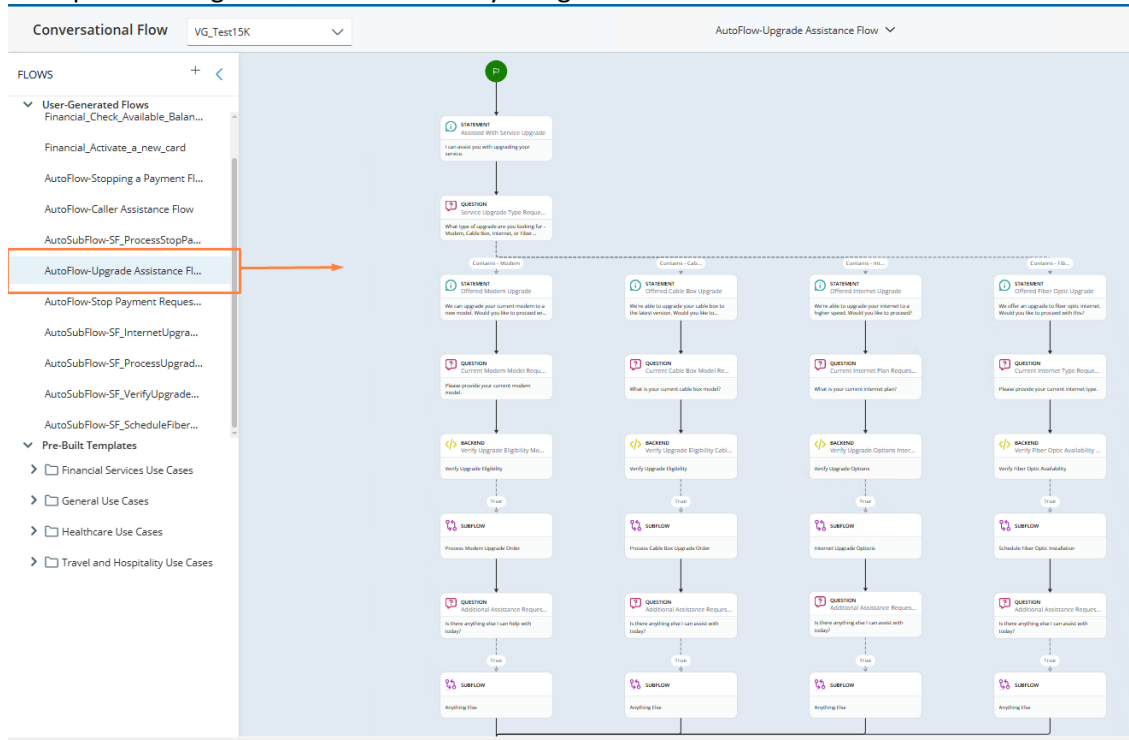


- Conversational Flow Page:
 - Users have three options to create a Flow.
 - a. New Flow
 - b. Import Flow
 - c. Automatic Flow Builder (Privilege Based)



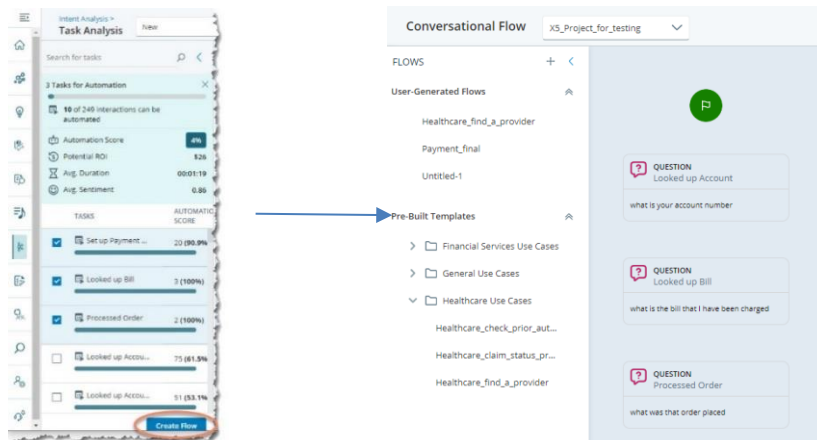
- **New Flow:** To create a Flow from scratch, the user can click this option.
- **Import Flow:** With this feature, the user can import entire conversational flows directly into the XO application using a simple, predefined file structure.
- **Automatic Flow Builder:** The user can now access the Automatic Flow Builder directly from both the Intent Analysis and Conversational Flow pages, to quickly generate flows for L3 intents. This feature is available only for users with the configuration parameter **Nexidia.XO.BOT** set to **Omilia** and with **Enlighten XO - Automatic Flow Builder** and **Enlighten XO - Export Data** privileges.

Example: A Flow generated automatically using LLM:



Task Analysis:

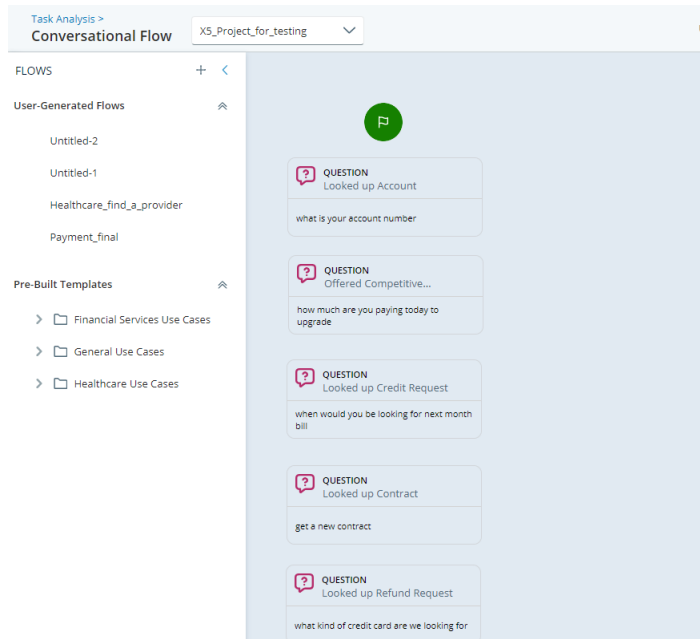
- In case one or more tasks are selected from the task grid because the user would like to create flows using these selected tasks. In this scenario, clicking on the “Create Flow” button will lead to the conversational flow, where for each selected task, one question action populated with the highest volume phrase for this action will appear. The user can continue to create the flow from that point.



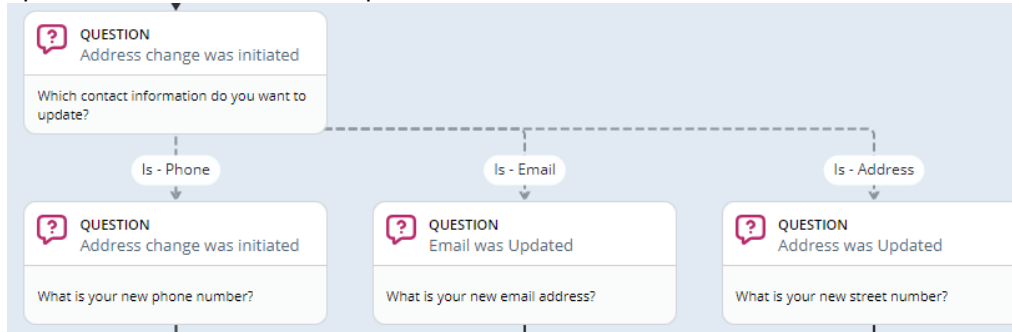
- In case the user chooses a specific task set from the main task analysis page, and wishes to skip the task analysis flow and go straight to the conversational flow, the user needs to click the context menu and from the options, hit create flow option, a create flow button will appear.



- In this scenario, clicking on the “Create Flow” button will lead to the conversational flow, where for each task on the selected task set, one question action populated with the highest volume phrase for this action will appear. The user can continue to create the flow from that point.



- Conditions:
 - The conversational flow allows the creation of a non-linear flow by providing the ability to add up to 6 conditions after each question. Each condition connector contains a label:



Starting from version 12.5 X9, users can now add conditions to Subflow and Backend blocks. This allows branching from Backend and Subflow actions based on their output variables, providing greater flexibility. For instance, you can now create branches depending on specific data returned by a Backend API or Subflow. Conditional connectors can originate from Question, Subflow, or Backend blocks, provided the block is fully completed.



- The variable type of the source of the conditioned connector determines the condition. The user selects the condition, for example, 'contains' and the value of the condition

Condition

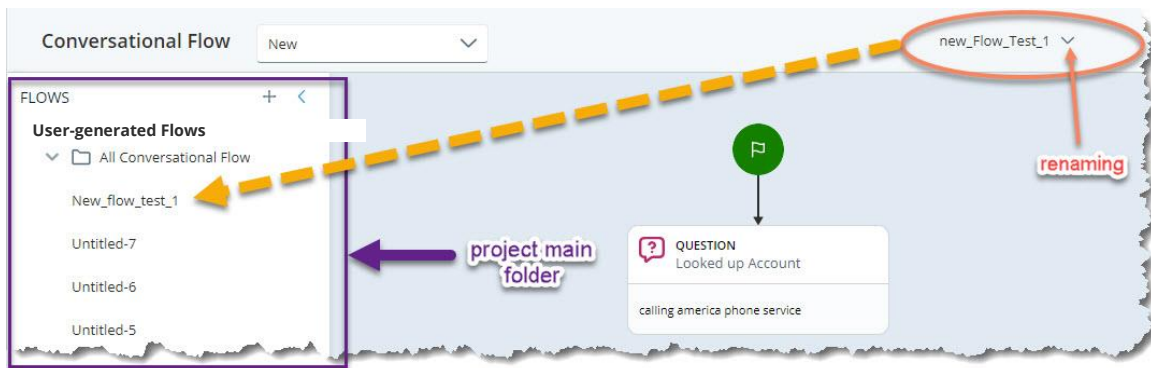
Select Condition

Contains

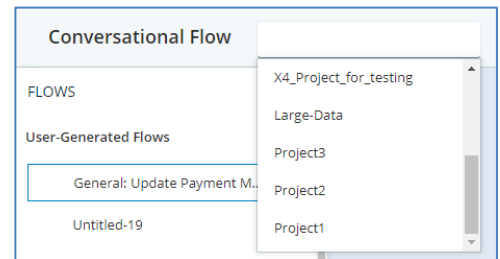
Enter Value

Debit

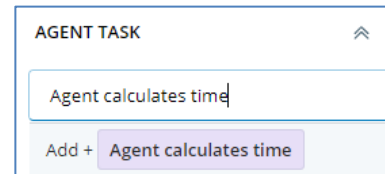
- Each flow is saved automatically, every minute, to the project's main folder. It can be renamed and loaded. The flow is also saved each time the user moves between flows, projects, or screens.



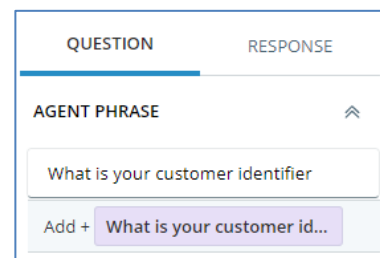
- Each project has a main folder where its saved flows exist. Navigating between the projects can be achieved by clicking on the project drop-down.



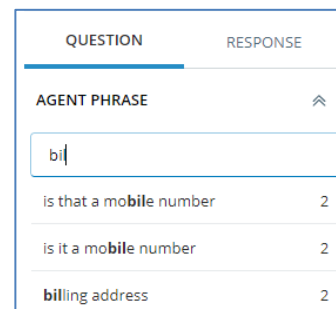
- Starting from X8, the user can change the task name for a specific action block. This will not result in a new task for other action blocks or flows within the project. In such a case, the user must provide agent and customer phrases to the Question and statement action types since no phrases will be linked to the added task.



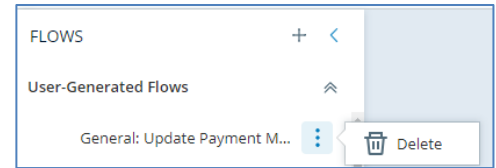
- Starting from X8, the user can ad hoc change or add a new phrase to the agent and customer's phrase. This action will not add the phrase to the list of phrases for the specific task, as displayed or exported from other areas of the Enlighten XO application, but rather is local to the flow/s to which it is added.



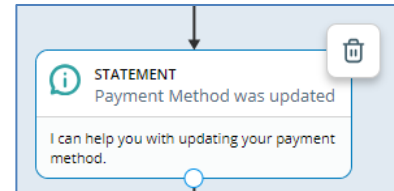
- Starting from X8, the user can search for agent tasks, agent phrases and customer phrases.



- Starting from X8, the user can delete flows, pointing to the three dots next to the flow name and clicking on the delete button:

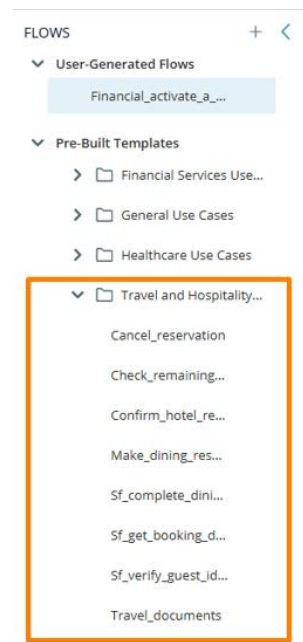


- Starting from X8, the user can delete action blocks using either right click on the block and select the delete icon or clicking on the 'delete' key. Pay attention that this action can create an orphan, causing a not completed flow.

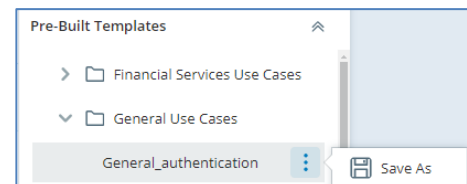


- Starting from X8, a new set of "Pre-Built Templates" have been added. There are three categories:
 - Financial Services Use Cases
 - General Use Cases
 - Healthcare Use Cases
 - Starting from version 12.5 X9, a new category called 'Travel & Hospitality' has been added.
 - Starting from version 12.5 X9, all the existing pre-built templates were updated with the relevant variable type.

Each category holds several flow templates that are ready to be used and exported as is or continue editing the flow for any need.

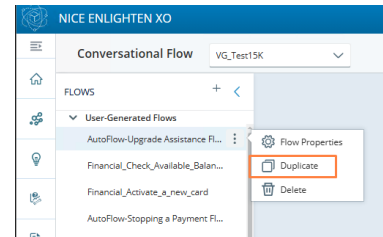


To export or edit the template, the user must click the "Save As" button. This action will create a copy of the template within the "User-Generated Flows" section.

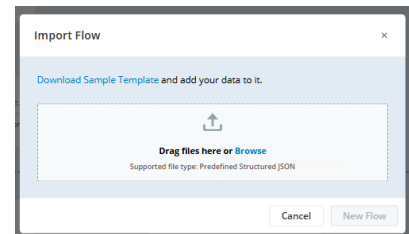


- Starting from X8, the exported file type will hold the date and time of the export, along with the flow name.
The file type is formatted depending upon the configuration of Nexidia.Xo.Bot within the system.

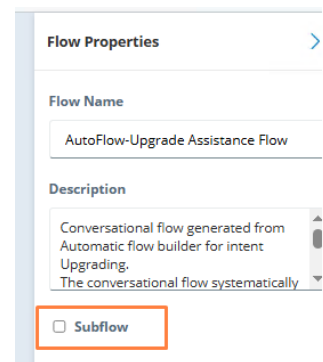
- Starting from X10, User will be able to duplicate an Existing Flow. This feature will enable users to quickly duplicate existing conversational flows, eliminating the need to build them from scratch.



- Starting from X10, users can import entire conversational flows directly into the XO application using a simple, predefined file structure. Whether it's a Main Flow or a Subflow, this feature gives you the freedom to design complex conversations in a clear and organized way.



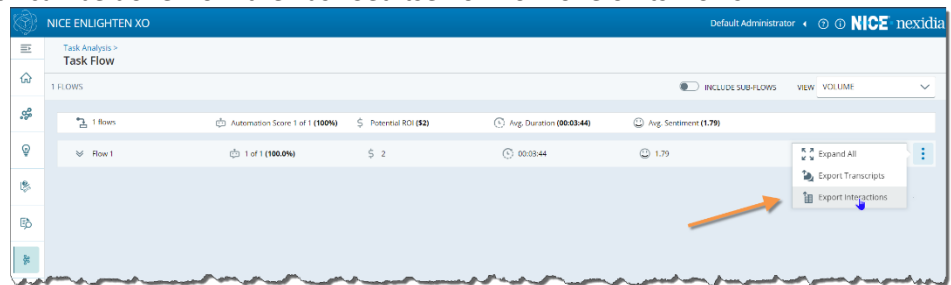
- Starting from X10, users can now turn any existing flow into a Subflow. This means it can be reused as a smaller, modular part of a larger flow—helping maintain consistency, improve scalability, and keep designs clean and organized.



- Conversational Flow limitations, known issues, and workarounds are noted in 'Known Issues/Behaviors and Workarounds' section.

Export Interactions List for Task Set/Flow

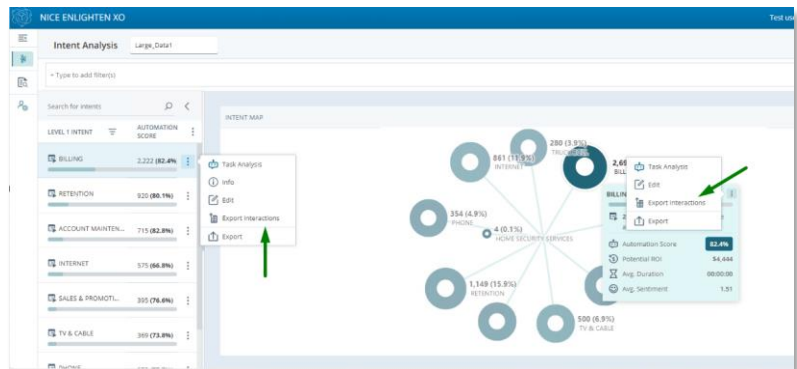
This functionality, added in v12.4 FP5.1 (September 2023) allows a user to export the list of media file names that match a given Task Set, which can be done from the Task Set itself or from one of its Flows.



As of v12.4 FP5.X X2 release (November 2023) the “INCLUDE SUBFLOWS” option allows users to view task flows within <selected Intent> that contain subsets of the selected Tasks. For example, if user has selected to View Flows for a Task Set containing three tasks, including sub flows would also display flows that contain only one or two of those three tasks.

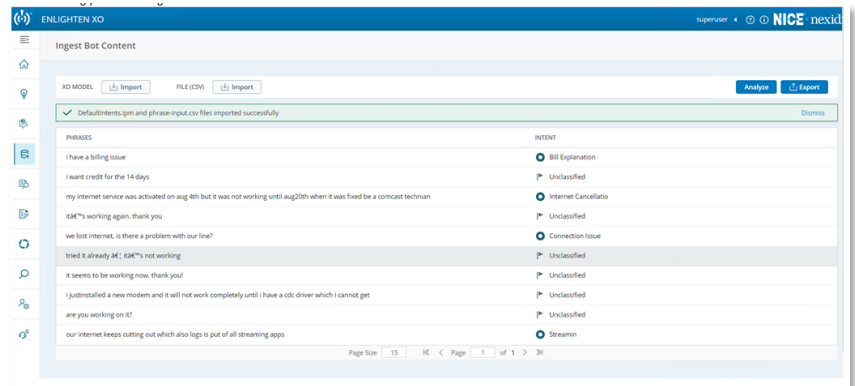
Export Interactions List for intents

This functionality, added in v12.4 FP5.2 X4 release (January 2024) allows a user to export the list of media file names that match a given Intent, which can be done via the 3-dots menu item as pictured below:



Bot Utterance Failure Analysis

Enlighten XO audits various user actions, as listed here. These actions are reportable in the Administration > Audit Report for Enlighten XO standalone customers, and in the Analysis & Reporting > Audit reports for customers with both Enlighten XO and Interaction Analytics.



User Activity Auditing

Enlighten XO audits various user actions, as listed here. These actions are reportable in the Administration > Audit Report for Enlighten XO standalone customers, and also in the Analysis & Reporting > Audit reports for customers with both Enlighten XO and Interaction Analytics.

- ◆ Enlighten XO - Activated Project
- ◆ Enlighten XO - Create Project
- ◆ Enlighten XO - Delete Project
- ◆ Enlighten XO - Edit Project
- ◆ Enlighten XO - Export Intent Data
- ◆ Enlighten XO - Export Task Data
- ◆ Enlighten XO - Hide/show Phrase
- ◆ Enlighten XO - Mark Intent as Work Complete
- ◆ Enlighten XO - Mark Task as Work Complete
- ◆ Enlighten XO - Played/Viewed File
- ◆ Enlighten XO - Rename Intent
- ◆ Enlighten XO - Rename Task
- ◆ Enlighten XO - Un-Mark Intent as Work Complete
- ◆ Enlighten XO - Un-Mark Task as Work Complete
- ◆ Enlighten XO - Updated ROI Parameter

As of X8, a new set of audit user actions were added for the conversational flow

- ◆ Enlighten XO - Create Conversational Flow
- ◆ Enlighten XO - Delete Conversational Flow
- ◆ Enlighten XO - Edit Conversational Flow
- ◆ Enlighten XO - Export Conversational Flow
- ◆ Enlighten XO - Rename Conversational Flow

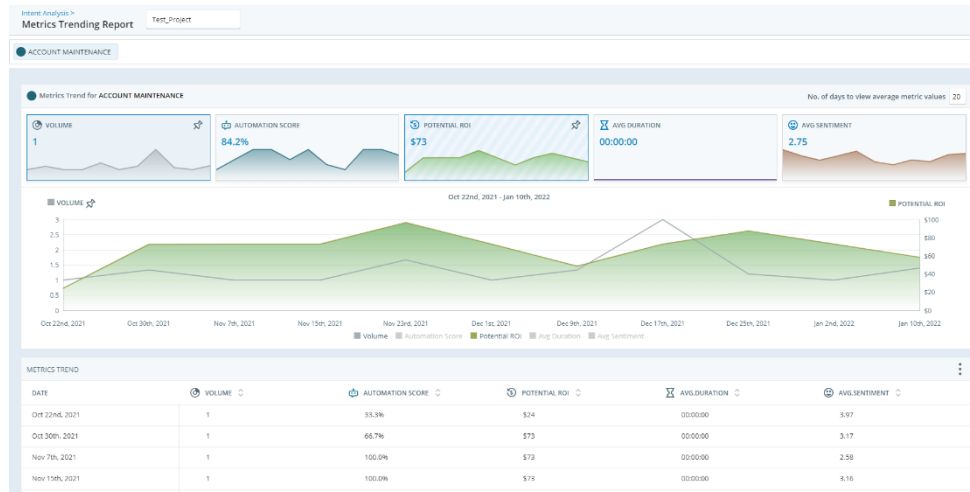
Enhanced Intent Trending

The v12.4 FP4 release adds enhanced trending to the Enlighten XO application, as described below.

Intent Trending: Compare Metrics

As noted previously, the left panel for Intent Analysis includes a widget for the Enhanced Trending Dashboard, which provides various trends for the selected Intent:

- Volume
- Automation Score
- Potential ROI
- Avg Duration
- Avg Sentiment



These metrics provide the user insight into how their automation efforts have impacted key metrics (e.g., has the volume for this intent decreased, do the interactions getting through to a live agent have shorter average duration, etc.).

The “Number of days...” field allows the user to indicate the time period (e.g., last <n> days) over which to calculate the average presented in the small tiles across the top of the screen.

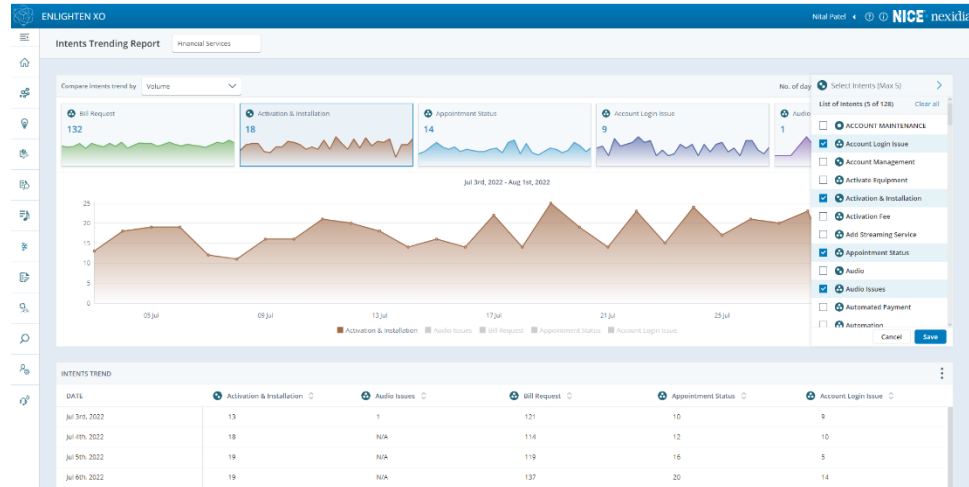
The user can compare various metrics (Volume, Automation score, ROI, Ave. Duration, Sentiment), two at a time, for a selected intent on a graph to view the trend for that intent. Simply click on the desired metrics to update the view on the larger trend chart.

The user can see (and export to CSV) the same trending data in the tabular format at the bottom of the screen.

Intent Trending: Compare Intents

This screen can be accessed from the Compare Metrics screen too via “Compare Intents” link on top right menu.

By default, the “Compare Intents” view is accessible via the main navigation item and displays the top 5 intents for the user’s selected (via drop-down list) metric.



The user can access all other available intents in the system from the dropdown list available in the top right menu from which the user can select any 5 of them at a time and see their trend on the main Compare Intents screen. The user can search for the intent using the search box in that dropdown box.

The user can compare any two intents on the larger trend chart by clicking on the corresponding small tiles across the top of the screen.

The “Number of days...” field allows the user to indicate the time period (e.g., last <n> days) over which to calculate the average presented in the small tiles across the top of the screen.

The user can see (and export to CSV) the same trending data in the tabular format at the bottom of the screen (via the “3-dots” menu item at the top-right of that grid).

Data Export via API

Version 12.4 FP4 enhances the existing data export capability by adding an API that can be used to request data from the Enlighten XO application. A Developers Guide is available in conjunction with the v12.4 FP4 release. Note that this API provides ‘same as’ functionality to match the manual export capability in v12.4 FP4.

User Interface Localization & Accessibility

The Enlighten XO user interface is currently available in English only. Full UI localization in all languages supported by Interaction Analytics and Quality Central is planned for late 2023.

Compatibility

This version of Enlighten XO is compatible with Interaction Analytics v12.5 FP5.1

This version of Enlighten XO processes interactions in English only.

Known Issues/Behaviors and Workarounds

Issue	Work-Around
In Conversational Flow, saved copies of pre-built templates from versions prior to version 12.5 X9 are not compatible with Omilia mini apps after upgrading to version 12.5 X9.	n/a - a future release will address this issue.
Refreshing the page during project creation may result in the project being saved without a model file.	Ensure model file is included before saving.
In Conversational Flow, flow generated by Automatic Flow Builder, if the task generated by the LLM is not available in the project, it will not be displayed.	In such cases, the user must manually select an alternative task from the project.
On the Task Analysis page, the Export dropdown will not display the Autoflow-Custom and Autoflow-Example options when the Nexidia.XO.BOT parameter is set to Amelia.	This issue will be resolved in a QF OTO X10.
In both imported and Automatic Flow Builder flows, removing a phrase from a task may not display the expected phrase list even when the task belongs to the current project due to missing task to project mapping.	n/a - a future release will address this issue.
Attempting to save a conversational flow from a prebuilt template while quickly loading another prebuilt flow may trigger an exception, causing both actions to fail.	To prevent errors, refrain from loading multiple prebuilt flows at once. Instead, load them one at a time.
In Conversational Flow, exporting a flow with a name that contains special characters results in an unhandled error.	Avoid using special characters in the flow name.
In Conversational Flow, when a user creates a large flow and exports it for specific supported bots, an alignment issue may occur where the blocks become misaligned due to X, Y coordinate discrepancies.	This issue will be resolved in a QF OTO X9.
The Enlighten XO project will not start processing if some interactions sampled from the specified Target Media Set have not been successfully transcribed.	Create your Target Media Set with a filter for overall Sentiment Score such that it will capture all interactions for which a score (and, therefore, a transcript) is available (e.g., Sentiment < 999).
NXQC-35809 - Exception occurs when there is a special character in the intent name, like *, and try to search them in the Intent search section	Remove special characters from the Project name.
Entity detection in Enlighten XO is known to have some limitations and is expected to exhibit some cases where entity detection is missed, or mis-identified (partial or incorrect words tagged as [<ENTITY>])	n/a – future releases will improve this capability.
Conversational Flow: If there is no customer response for an agent phrase, the customer response field will be blank.	The user needs to either choose a different agent phrase or select a different task.
Conversational Flow: When preparing the conversation flow on the canvas, there will be situations where the action blocks overlap.	The user can drag and move actions to adjust them as desired.

Issue	Work-Around
Conversational Flow: All hidden phrases and customer responses will still appear in the conversational flow.	n/a - a future release will address this issue.
Conversational Flow: Multiple users can simultaneously work on the same flow, which might lead to incorrect or faulty flows that the user does not desire.	n/a - a future release will address this issue.
Conversational Flow: In some chatbots cyclic conversational flows cannot be rendered due to the chatbot limitation.	The user needs to make sure to adjust the flow when building it so that no cyclic flow is created.
Conversational Flow: If multiple conditional inputs are used for the same action, the labels might overlap, making both texts visible in rare cases.	n/a - a future release will address this issue.
Conversational Flow: When multiple conditional inputs to the same action come from different entry points (left or top), clicking one condition label may hide the label from the other entry point.	n/a - a future release will address this issue.
Conversational Flow: In some chatbots, if there are two (or more) connectors with the same source and destinations, the rendered flow on the chatbot will hold only one connector to the destination.	The user needs to use a single condition with the operator "One of the following options", instead of creating two separate conditioned connectors.

Key Resolved Issues

Issue
NXQC-55316: Fixed an issue in the logic to make sure Conversational Flow shows validation error on incorrect input values while adding conditional values.
NXQC-57585: Fixed an issue in the XO > Task Flow page to make sure that the flows do not overlap with each other and the arrows between the flows appear correctly.
NXQC-57730: Fixed a performance issue to ensure that Conversational flow is generated within a short period of time, for a large XO project.
NXQC-41764: With this fix, all previously completed projects will now work with trending data.
NXQC-41607: The number of pages displayed on the Intent Export page is as per the records fetched after applying various filters.
NXQC-41523: Now if a project is deleted from Experience Optimization (XO), you will not see its data on any XO page panels.
NXQC-41451: The dropdown menu will now only display the list of projects that have been successfully processed with a status marked as 'Done'.
NXQC-41336: Intent/Task Analysis: Now the scrollbar in the info/Show phrase panel will not hide itself when dealing with large data volumes.
NXQC-41294: After filters are applied on the Export Intent page, the order of pages displaying the fetched records is maintained correctly.
NXQC-41219: The Hide phrase functionality on the Intent Analysis > Info panel hides the phrases for an intent.
NXQC-40738: Performed minor user interface improvements on the Intent Analysis page.
NXQC-40735: Performed minor user interface improvements on the Task Analysis > Task Flow preview page.

Issue
NXQC-40716: If the Experience Optimization (XO) Project creation fails due to model file not uploaded correctly, the 'Activate Trending Report' toggle button is turned off.
NXQC-40552: If the user session times out, the user is logged out from the Experience Optimization (XO) application.
NXQC-40515: If there are no sub-flows available for a task, then a message 'Export Task Flow page - No additional task flows found for the selected task flow' is displayed if the user selects the 'Include Sub-flow' toggle button on the Task Analysis > Task Flow page.
NXQC-40513: Now filter will retain on task analysis page while coming back from task set export or individual task export page
NXQC-40511: Performed minor user interface improvements on the Task Flow > Export Task Flow preview page.
NXQC-40494: External users should be able to get responses from the XO Data API Service.
NXQC-40465: The flows displayed on the Task Flow page are sorted in ascending order of Automation Score.
NXQC-39192: For any given Intent, the Average sentiment value is displayed up to 2 decimal places in the Intent Analysis > Info section as well as in the JSON file exported using 'Intent Analysis > Export Project Data' option.
NXQC-43737: Users experienced difficulty accessing chat transcripts in XO Standalone, but they will now have the capability to view chat transcripts within the existing XO playback screen across all XO configurations.
NXQC-33327: Interactions that fail during data processing will not be included in trend analysis.
The following issues were resolved in the X7 release (June, 2024):
NXQC-47575: Fixed issues in Project Processing.
NXQC-46976: With this fix, customers will be able to see the interaction transcript for the chat project in XO Standalone.
NXQC-46975: With this fix, the customers will be able to see the transcript for the project in XO Standalone.
NXQC-46534: Fixed an issue in the UI to render phrase details in the Intent Analysis page while switching between projects.
NXQC-46068: With this fix, the customers will be able to see the correct data when they apply the Automatable Filter.
NXQC-46066: Fixed cosmetic issues on the Task Analysis page.
NXQC-46058: Now users can see the original name of the media instead of the Interaction ID on the show interaction page.
NXQC-46005: Support XO to view chat transcript without using Text Grid
NXQC-45377: Selected intent was shifting to parent intent while adding data set bar filters. This issue has been resolved.
NXQC-45128: With this fix, the user will be able to use the newly created TMS after creating 10 other TMS
NXQC-44659: With this fix, the user will see tabs related only to XO Standalone.
NXQC-44303: Fixed this cosmetic issue on the Intent Analysis page.
NXQC-44211: With this fix, the Intent Map will display colors corresponding to the filter selected in the Data Layer Filter panel on the right
The following issues were resolved in the X8 release (September, 2024):

Issue
NXQC-5118: Intent Process used to fail when the system had no internet. A new version of IntentProcessor.exe has been included in X8 version with the fix that it will work on a system with no internet access.
NXQC-4868: With this fix, the Avg. Duration filter on the Task Analysis page will function correctly when used in combination with another data layer filter.
NXQC-4849: Resolved the issue to ensure Agent Tasks in the Conversational flow (Question/Statement/Backend action) are uniquely identified, even when they contain identical text.
NXQC-48439: Conversation Flow - The flow name should automatically trim white spaces. Updated the flow name logic to remove multiple spaces between words and prevent the acceptance of a blank name.
NXQC-48425: With this fix, the Customer Response section in the Conversational Flow's action block will display the message 'No data is available for the selected agent phrase' when there is no associated Customer response for the chosen agent phrase.
NXQC-46004: Implemented a fix to display the 'Work Completed' icon when searching for tasks on the Task Analysis page.
NXQC-45996: The list of Intents in the left panel of the Intent Analysis page is properly refreshed and displayed in order of volume when an Intent is hidden or made visible.
NXQC-45824: Resolved the issue where the selected project's data on the Intent Analysis page was not retained when users navigated between different pages within the XO application
The following issues were resolved in the X9 release (February, 2025):
NXQC-51973—Conversational Flow: We fixed the issue where the "OK" button remained disabled until the user added, typed, and selected a customer response.
NXQC-51188 - Conversational Flow: Agent tasks in the Pre-Built Template are now correctly mapped to the IPM file, ensuring alignment between template tasks and those in the IPM file.
NXQC-51577 – Conversational Flow: A warning message is now displayed when a user converts a Question Action block to Escalation or Endflow, alerting them that this action will orphan the connected blocks or connectors.

Appendix A: Summary of Features per Release

The following table provides a listing of the key features in each Enlighten XO release starting with v12.4 FP3, in Q4 2022.

Release	Date	Features
V12.5 FP1 X10	July 2025	• Conversational Flow – Automatic Flow Builder Introduced an Automatic Flow Builder powered by LLM for Omilia-configured XO applications. This feature auto-generates conversational flows, significantly reducing manual design effort and providing teams with an instant visual starting point to understand dialogue structures. • Conversational Flow – Duplicate User-Generated Flows Added the ability to quickly duplicate existing user-created flows, eliminating the need to recreate them from scratch. • Conversational Flow - Flow descriptions and subflow settings Added the ability to add Flow Descriptions and link subflows to other flows to help reuse flows and improve export for Omilia. • Conversational Flow – Import Flows Directly Users can now import conversational flows directly into the Conversational Flow page using a predefined file structure, with the flexibility to import as either a Main Flow or a Subflow. • Conversational Flow – Convert Flow to Subflow Introduced an option to convert any flow into a subflow seamlessly. • Conversational Flow – Added “Anything Else” action block specifically designed for Omilia-based flows to let bots ask users if they have additional intents. • Task Analysis – Export Interactions Users can now export key interaction data for the entire task set directly from the Task Analysis page. • Intent Export Page – Recommended Phrases Launched a new advanced algorithm that recommends the most impactful phrases for Intents based on semantic distance and intelligent grouping techniques. • Integration with Text Analysis Satellite V5.1 Integrated XO with Text Analysis Satellite V5.1 via Voice Grid, significantly enhancing result accuracy with the powerful 5.1 LLM engine.
V12.5 x9	Feb 2025	• Conversational Flow - Flow builder now supports additional variable types. • Conversational Flow - Supports 2 additional agent rephrased prompts on the question action, or specific supported bots. • Conversational Flow - Implementation of Data Privacy for Question, Statement and Backend action type or specific supported bots • Conversational Flow - Capability to branch using the condition connector from Subflow and Backend Action Type. • Conversational Flow - Addition of a new pre-built template flows Folder "Travel and Hospitality" • Conversational Flow - Enhancement of existing Pre-Built Templates by mapping the actions to support the supported bots

V12.4 FP5.2 X8	Sept 2024	• Enlighten XO – Now supports a new chatbot integration. • Create a New Project – Users creating a Project can now select an intent model from a pre-defined common folder on the server, located in the shared Enlighten XO directory. • Conversational Flow – New action blocks have been added named "Sub-Flow," "End Flow," and "Escalation." • Conversational Flow – Users can now delete an entire flow or individual blocks within a flow. • Conversational Flow – Users can quickly load a pre-built flow from a selection of options, complete with agent and customer phrases for each statement and question. The ready-to-use flow can be easily customized to meet specific needs. • Conversational Flow – Users can now search for or add Agent Tasks, Agent Phrases, or Customer Responses. • Conversational Flow – Users can view applied conditions as labels on the connectors, making it easier to follow and navigate the flow. • Conversational Flow – Users can now access user activity / Audit results related to the Conversational Flow via the Audit Reports.
V12.4 FP5.2 X7	July 2024	• Conversational flow – Build and export flow using a set of action types that serve as a toolbox and incorporate the XO agent tasks and phrases. • Project Administration – Support the dual topic model (IPM)
V12.4 FP5.2 X6	April 2024	• Filters panel – allow direct numeric text entry into fields (in conjunction with sliders) • Interaction Playback – additional data for each interaction
V12.4 Fp5.2 X5	March 2024	• XO player – Export <current interaction> Transcript as text • Task Flow Export – add media type of exported file to the export file name • Task Analysis – “Find task” (search for task by name) added to task list • Intent Export – ability to merge level3 Intents to facilitate grouping of numerous different Intents into a common path (e.g., send to live agent) • Intent Analysis – ability to hide Intents from the UI and data exports
V12.4 Fp5.2 X4	January 2024	• Intent Export – Filter phrases by a free text • XO can now retain the last visited project, so the next time that an end-user logs in, the system will show the project that was previously used • Intent Analysis - Users will be able to download the list of interactions grouped under each intent, facilitating a clear understanding of the specific calls and chats associated with each intent • Project Administration – now includes the ability to view the number of interactions that failed during the project run, per failed reason. This feature also allows download of a list of those interactions per reason
V12.4 Fp5.1 X3	December 2023	• Task Analysis – now includes a “Volume” column enabling users to see how many interactions contained each set of Tasks • The phrase viewer (in Intent Analysis and Task Analysis) will now support infinite scroll (loading 50 items at a time) • Intent Export updates: ○ When exporting to AutoFlow, the intent preview will sort by Volume by default ○ Intent Export now includes an “INCLUDE” column of checkboxes

		○ Contains an option (filter: Phrase Format) for Phrase vs. Snippet ○ System automatically removes duplicate phrases (if they occur across multiple <i>different</i> intents in the export) ○ A new configuration for 'clean up' of exported data, which allows administrators to indicate which artifacts or words to remove from transcript view/export (e.g., "unk", "...", etc.). Configuration value is Nexidia.XO.PhraseCleanupPatterns • Enlighten XO will retain the last project visited for a user within a given user session
V12.4 Fp5.1 X2	November 2023	• Intent Export Page – Filters Panel • Intent Export – Filter by Volume (%), Automation Score • Task Set/Flow – Toggle data • XO new feature Toggle capability (not customer-facing)
v12.4 FP5.1 X1	October 2023	• Intent Export – Word Count filter, addition of all (three) Intent hierarchy levels to the Intent CSV export (in all cases). • Intent Analysis – option to Export ALL Intents • Hard delete XO projects/data (previously had been soft delete)
v12.4 FP5.1	September 2023	• Intent Analysis – Option to Export JSON Data for entire Enlighten XO Project • Task Flows – New AutoFlow – Custom option allowing configuration of conversation to export as JSON format for direct import into SmartAssist – Digital Employee Builder. Prior AutoFlow option renamed as "AutoFlow – Example" • Task Set and Task Flow – Option to "Export Interactions" media file names to CSV
v12.4 FP5	August 2023	• Upgrade without impacting existing custom .pred file • Export of Intent Analysis "Dashboard" view (Intents & Metrics) to CSV • Handle 6 different types of entities during the XO export to meet compliance requirements (NAME, LOCATION, NUMBER, CURRENCY, EMAIL, and ADDRESS) • Configure and preview data before exporting them from Intent Analysis • Configure and preview data before exporting them from Task Sets (Task Analysis) • Configure and preview data before exporting them from Individual Tasks • Configure and preview data before exporting them from Task Flows • Export interaction representing selected Task Flow in JSON format
v12.4 FP4	March 2023	• Updated the network diagram on the Intent Analysis with new design of nodes • Enhanced Intent Trending to compare different intents and its trends over time • Enhanced Intent Trending to compare different metrics for a selected Intent and its trends over time • Enhanced DataSet filters on the Intent Analysis screen to filter data based on Media Type (Audio or Chat), or Automatable (yes / no). • Export XO data via API • Export up to 10 transcripts of the conversation flows from the Task Flow screen (Released as FP4.2) • Added customer responses in the XO Task data exports • Select any intent from the XO project to compare on Intent Trending • Redact numbers before exporting the XO data (Released as FP4.2)

		• UI widget (checkbox) to select/deselect individual tasks on the Task Analysis page.
v12.4 FP3	November 2022	• Push XO primary intents to the NIA for Analytics purposes • Search across all the Intent levels on the Intent dashboard • GDPR and CCPA compliance • Trending report over time on the left panel of Intent Analysis screen • Enhanced version of the Ingest Bot Content